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Research Article

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第 1 页中文翻译 / 说明

《钢琴的组件模态综合时域仿真》

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研究论文 / 预印本

关键词：钢琴；物理建模；时域仿真；振动声学；组件模态综合（CMS）；有限元法（FEM）；快速多极边界元法（FMBEM）；非线性特征值分析；几何非线性；制造残余应力。

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Time-Domain Simulation using Component Mode Synthesis for a Piano

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Abstract

We present a physics based, time-domain numerical simulation framework that treats a full-scale grand piano as a comprehensively coupled vibroacoustic system, encompassing nonlinear hammer-string interaction, geometrically nonlinear and lossy strings with anisotropic end supports, a three dimensional orthotropic wooden body with structural damping, and exterior acoustic radiation. The approach combines hybrid interface component mode synthesis (CMS) and Galerkin method for the strings with a finite element (FE) body model and a fast multipole boundary element method (FMBEM) for the surrounding air. The large coupled body-air subsystem is reduced via a complex mode CMS based on nonlinear eigenvalue analysis, yielding complex modal parameters at string support locations together with transfer functions to virtual microphones for efficient auralization. Manufacturing-induced residual stresses are explicitly incorporated through nonlinear static analysis of the assembly process, enabling system level causal links from design and fabrication parameters to radiated sound. To our knowledge, this work represents the first fully integrated time-domain piano simulation that simultaneously couples: (i) a nonlinear, double layer generalized Maxwell model of hammer felt; (ii) a spatial, bidirectionally coupled string model with stiffness, dispersion, and zig-zag/anisotropic boundary conditions at bridge and bearing; and (iii) a 3D body-air model reduced by complex modal synthesis, including residual stress effects. The method reconstructs driving point mobilities and perceptually salient phenomena—partial dispersion, polarization exchange, and “ringing”—and provides audible examples as supplementary material. Beyond piano acoustics, the pipeline constitutes a generalizable methodology for large scale, strongly coupled nonlinear structural-acoustic simulations in engineering design, enabling parameter sensitive, time-domain auralization with tractable cost.

摘要

本文提出一种基于物理的时域数值仿真框架，将一台全尺寸三角钢琴作为一个完整耦合的振动声学系统处理。系统包括：非线性击锤-弦相互作用；具有几何非线性和损耗、且端部支承各向异性的琴弦；具有结构阻尼的三维正交各向异性木质琴体；以及外部声辐射。

对琴弦，方法结合混合界面组件模态综合（CMS）与 Galerkin 法；琴体采用有限元（FE）模型；周围空气采用快速多极边界元法（FMBEM）。大型琴体-空气耦合子系统通过基于非线性特征值分析的复模态 CMS 降阶，得到弦支承位置的复模态参数，以及到虚拟麦克风的传递函数，以便高效可听化。

制造过程产生的残余应力通过装配过程的非线性静力分析显式纳入，从而建立从设计/制造参数到辐射声音的系统级因果链。作者认为，这是首个在统一时域框架中同时耦合以下三项的完整钢琴仿真：（i）击锤毡的双层非线性广义 Maxwell 模型；（ii）具有刚度、色散、琴桥/弦枕 zig-zag 与各向异性边界条件的空间双向耦合弦模型；（iii）包含残余应力影响、并经复模态综合降阶的三维琴体-空气模型。

该方法可重建驱动点导纳，并重现分音色散、偏振能量交换和“ringing（金属鸣响）”等听觉上重要的现象。

Keywords: piano, physical modeling, time-domain simulation, vibroacoustics, component mode synthesis (CMS), finite element method (FEM), fast multipole boundary element method (FMBEM), nonlinear eigenvalue analysis, geometrical nonlinearity, manufacturing-induced residual stress

1 Introduction

Achieving an ideal piano tone requires identifying an optimal combination of design parameters across the entire instrument, which constitutes a highly complex, large-scale vibroacoustic system. Traditional empirical approaches—heavily dependent on craftsmanship, experience, and iterative prototyping—are often inefficient due to the large number of interacting variables and unavoidable uncertainties arising from natural material variation and manual adjustment. These limitations make it difficult to systematically evaluate how specific structural and acoustic factors influence the resulting sound. To address this challenge, physics-based modeling and numerical simulation have emerged as promising tools for understanding and predicting the mechanisms of piano sound production.

A substantial body of research on physical modeling of piano components has been accumulated over several decades. Early studies focused on simplified interactions such as the coupling between a linear string and a linear bridge, solved via analytical techniques including Laplace transforms [1]. Subsequent work addressed the nonlinear hammer–string interaction [2], the hysteretic behavior of hammer felt [3], and the generation of longitudinal string vibration [4]. More recently, high-fidelity time-domain simulations have incorporated two-dimensional and three-dimensional models of the soundboard and surrounding air [5]. While each of these studies contributed valuable insights into individual subsystems, none provided a complete representation of the piano as an integrated acoustic instrument. In particular, previous approaches often simplified essential elements such as geometrical nonlinearity, string end-support anisotropy, wood orthotropy, manufacturing-induced residual stresses, and the three-dimensional coupling between the body and exterior acoustics.

From the perspective of piano design and engineering practice, ignoring these factors is not acceptable when the goal is to clarify the causal relationship between design specifications and the resulting sound. Based on the authors’ experience in piano development, the following factors (a)–(i) are indispensable for accurate sound prediction:

- (a) nonlinear viscoelastic compression of hammer felt;
- (b) spatial string vibration including stiffness, losses, and geometrical nonlinearity;
- (c) rear duplex segments;
- (d) three-dimensional complex orthotropic elasticity of wooden components;
- (e) inclusion of nearly all structural elements, such as soundboard, ribs, bridge, rim, key bed, cast-iron frame, legs, casters, and top lid;
- (f) manufacturing-induced residual stresses arising from successive gluing, bolting and stringing processes;

1 引言

要获得理想的钢琴音色，需要在整件乐器这一高度复杂、尺度巨大的振动声学系统中寻找设计参数的最佳组合。传统经验方法高度依赖工艺、经验和反复制造样机；大量相互作用变量，以及天然材料差异和人工调整带来的不确定性，使这种方法效率有限，也难以系统判断某个结构/声学因素如何影响最终声音。基于物理的建模和数值仿真因此成为理解并预测钢琴发声机制的重要工具。

过去数十年已经积累大量组件级物理建模研究：早期研究线性弦与线性琴桥耦合；随后涉及非线性击锤–弦相互作用、击锤毡迟滞、纵向弦振动的产生；近年还出现包含二维/三维响板与周围空气的高保真时域仿真。然而这些研究多数针对孤立子系统，没有完整表示钢琴作为一件整体声学乐器。过去方法尤其常简化几何非线性、弦端支承各向异性、木材正交各向异性、制造残余应力，以及琴体与外部声场的三维耦合。

根据作者的钢琴研发经验，为准确预测声音不可缺少：(a) 击锤毡非线性黏弹压缩；(b) 包含刚度、损耗与几何非线性的空间弦振动；(c) 后弦段；(d) 木质部件三维复杂正交各向异性弹性；(e) 尽可能完整的结构部件；(f) 胶合、螺栓固定和上弦产生的制造残余应力。

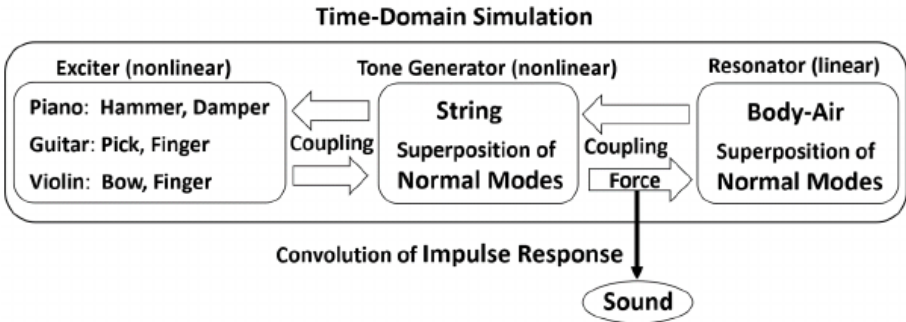


Fig. 1 Block diagram of the proposed sound-synthesis framework for string instruments (e.g., piano, guitar, and violin).

- (g) three-dimensional body–air coupling with non-proportional damping;
- (h) anisotropic end-support conditions of strings; and
- (i) acoustic radiation from all surfaces of the piano into an open field.

A piano that incorporates all of these factors forms a highly intricate system, characterized as a three-dimensional coupled structure involving nonlinear subsystems (hammers and strings) and a large-scale linear vibroacoustic subsystem (body and air). Direct time-domain simulation of such a system is computationally prohibitive due to its immense dimensionality. To overcome this challenge, we employ an advanced CMS-based reduction strategy that enables efficient treatment of large-scale exterior vibroacoustic problems—an application of CMS far less explored than conventional interior structural dynamics [6].

Depending on the characteristics of each substructure, we adopt appropriate CMS formulations (free-interface, fixed-interface, and hybrid-interface) to model interactions among the strings, body, and surrounding air. The normal modes of the coupled body–air system are obtained through a nonlinear eigenvalue analysis combined with the fast multipole boundary element method (FMBEM), and are orthogonalized into complex modes for time-domain use. These reduced modal parameters allow us to synthesize sound by convolving the impulse response between string support points and virtual microphone positions with string forces computed in the time domain. The block diagram of the proposed sound synthesis method is shown in Fig. 1. Notably, this block diagram differs crucially from the well-known block diagram of [7]: the traditional block diagram ignores the feedback from the linear resonator (body–air system) to the strings, whereas ours takes this feedback into account. As shown in this figure, the basic idea of the presented method can be applied to other musical instruments, such as guitars and violins. These stringed instruments share the property that they can be modeled as a coupled system consisting of nonlinear exciters (such as hammers, a pick, or a bow), nonlinear strings, a linear body, and linear air.

The remainder of this paper is organized as follows. Section 2 presents the physical models and governing equations of each subsystem and the fully coupled system. Section 3 describes the numerical procedures, including eigenvalue analysis, modal reduction, and time-domain integration. Section 4 provides simulation results with

引言 (续)

还必须考虑: (g) 具有非比例阻尼的三维琴体–空气耦合; (h) 弦端各向异性支承; (i) 钢琴全部表面向开放声场的辐射。

包含上述因素后, 钢琴成为由非线性击锤/琴弦子系统与大尺度线性琴体–空气子系统组成的复杂三维耦合结构。直接做全系统时域仿真在计算上不可承受, 因此作者采用高级 CMS 降阶策略。根据各子结构特征, 分别使用自由界面、固定界面和混合界面 CMS。琴体–空气耦合正规模通过结合 FMBEM 的非线性特征值分析得到, 再正交化为适用于时域的复模态。

这些降阶参数使系统可以把弦支承点到虚拟麦克风的脉冲响应, 与时域计算得到的弦端力进行卷积来合成声音。图 1 的框图与经典文献 [7] 有关键区别: 传统框图忽略线性共振器 (琴体–空气系统) 反馈到弦的作用, 而本文保留该反馈。作者指出这一思想同样可用于吉他、小提琴等弦乐器。

全文结构: 第 2 节给出各子系统及完全耦合系统的物理模型与控制方程; 第 3 节介绍特征值分析、模态降阶与时域积分; 第 4 节给出仿真与可听化结果; 第 5 节讨论物理意义; 第 6 节总结并讨论未来工作。

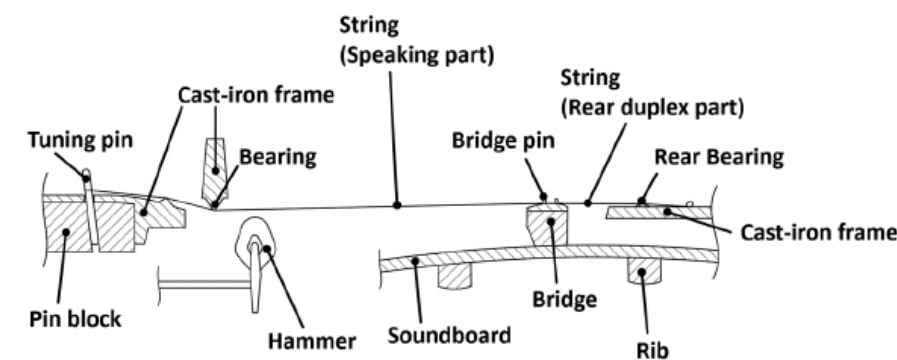


Fig. 2 Schematic diagram of the main structural components of a grand piano.

audible examples, and Section 5 discusses the physical implications of the findings. Section 6 concludes the study and outlines possible directions for future work.

2 Model and governing equations

First, we define the coordinate axes. For the object coordinates (the global coordinates) of the body, the right-handed Cartesian coordinates (X, Y, Z) are used. In the case of a grand piano, the XY -plane is usually parallel to the floor on which the piano is placed, with the X -axis from the bass side to the treble side, the Y -axis from the player side to the piano side, and the Z -axis vertically upward. The strings of a grand piano have the speaking, rear duplex, and other parts, as shown in Fig. 2. Notably, herein, the “string” includes only the speaking and rear duplex parts, and the remaining parts are included in the “body”. For the object coordinates (the local coordinates) of the string at rest, the right-handed Cartesian coordinates (x, y, z) corresponding to each part are used. The x -axis aligns with the chord in the string-axis direction, the y -axis is set parallel to the XY -plane, and the z -axis is almost vertically upward. In the speaking part, the x -axis is defined as $x = -l^{[1]}$ at the end-support on the cast-iron frame, i.e., bearing (including agraffes) and $x = l^{[1]}$ at the end-support on the bridge. The y -axis goes from the treble side to the bass side. Conversely, in the rear duplex part, the x -axis is defined as $x = -l^{[2]}$ at the end-support on the cast-iron frame, i.e., rear bearing and $x = l^{[2]}$ at the end-support on the bridge, and the y -axis goes from the bass side to the treble side. Here, $l^{[i_L]} = L^{[i_L]}/2$. $L^{[i_L]}$ ($i_L = 1, 2$) are the lengths of the parts of the string at rest, where $i_L = 1$ and $i_L = 2$ denote the speaking and rear duplex parts, respectively.

2.1 Hammer

Pianos produce a soft tone that contains almost no high-frequency components when struck lightly and a powerful tone that contains many high-frequency components when struck strongly. This is mainly due to the nonlinear viscoelasticity of the hammer felt.

2 模型与控制方程

首先定义坐标。琴体采用右手全局坐标 (X, Y, Z) ： XY 平面通常平行于地面， X 从低音侧指向高音侧， Y 从演奏者侧指向钢琴内部， Z 竖直向上。琴弦含有效弦段 (speaking part)、后弦段 (rear duplex) 及其他弦段。本文所谓 “string” 只包括有效弦段和后弦段，其余弦段并入 “body”。

弦静止时采用局部右手坐标 (x, y, z) 。 x 沿弦轴， y 平行于全局 XY 平面， z 近似竖直。有效弦段在铸铁板/弦枕端定义 $x=-1^{[1]}$ ，琴桥端定义 $x=+1^{[1]}$ ；后弦段类似地使用 $1^{[2]}$ 。 $L^{[i_L]}$ 为相应弦段静止长度， $1^{[i_L]}=L^{[i_L]}/2$ 。

2.1 击锤

轻击钢琴时声音柔和且高频很少；强击时声音强劲并包含大量高频。主要原因是击锤毡的非线性黏弹性。

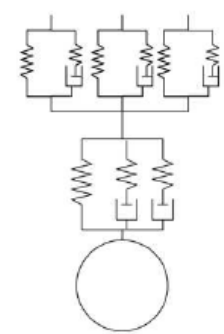


Fig. 3 Hammer model consisting of a lumped mass and a double-layer nonlinear generalized Maxwell model ($I_W = 3, I_{Hb} = 2$)

In viscoelastic theory, the Maxwell and Voigt models are the most basic. The former comprises a spring and a dashpot connected in series, whereas the latter comprises a spring and a dashpot connected in parallel. The generalized Maxwell model, which combines one spring and one or more Maxwell models in parallel, is a practical model for expressing linear viscoelasticity [8]. Furthermore, by replacing the compression w (where w is a non-negative real number) in the expression of this linear model with w^p (where p is a positive real number), the nonlinear generalized Maxwell model can be obtained. The hammer felt model with nonlinear hysteresis proposed by Stulov is a type of nonlinear generalized Maxwell model [3]. This paper proposes a double-layer nonlinear generalized Maxwell model that more accurately represents the dynamic compression characteristics of hammer felt (Fig. 3). The upper and lower layer of this model represent the viscoelasticity of “the hammer tips that contact each string” and “the hammer core”, respectively.

When using the lumped mass model with the assumption that the hammer moves on one axis, the equation of motion of the hammer is expressed as follows:

$$M_H \ddot{u}_H(t) = -f_{Hb}(t) \tag{1}$$

where M_H is the hammer mass, $u_H(t)$ is the displacement of the hammer mass, and $f_{Hb}(t)$ is the force exerted by the hammer mass on the lower layer of the hammer felt. t is time, and an overdot denotes differentiation with respect to time. The equation of equilibrium on the joint point of the upper and lower layers is expressed as follows:

$$f_{Hb}(t) = \sum_{i_W=1}^{I_W} f_{Ha}^{[i_W]}(t) \tag{2}$$

where $f_{Ha}^{[i_W]}(t)$ is the force exerted by the upper layer of the hammer felt on the string, and i_W and I_W are the index and number of strings corresponding to a certain key, respectively. When using the nonlinear generalized Maxwell model with the assumption that the hammer felt deforms only in the direction of the hammer motion, the

2.1 击锤（续）

Maxwell 模型由弹簧与阻尼器串联组成，Voigt 模型由二者并联组成。广义 Maxwell 模型把一个弹簧与一个或多个 Maxwell 支路并联，是表达线性黏弹性的实用模型。如果把线性模型中的压缩量 w 替换为 w^p ($p>0$)，即可得到非线性广义 Maxwell 模型。Stulov 提出的带非线性迟滞的击锤毡模型就是其中一种。

本文提出双层非线性广义 Maxwell 模型（图 3），用于更准确表示击锤毡的动态压缩特性。上层表示“分别接触各根弦的锤尖区域”的黏弹性；下层表示“击锤核心”的黏弹性。

击锤本体采用集中质量，并假定沿单一轴线运动。式（1）： $M_H \ddot{u}_H(t) = -f_{Hb}(t)$ 。其中 M_H 为击锤质量， u_H 为击锤质量块位移， f_{Hb} 为击锤质量块施加到下层毡上的力。上下层连接点满足力平衡：式（2）中，下层力 f_{Hb} 等于同一琴键所有弦对应的上层接触力 $f_{Ha}^{[i_W]}$ 之和。 i_W 是弦索引， I_W 是该键的弦数。

force-compression relational expressions for the upper and lower layers of the hammer felt are given as follows:

$$f_{Ha}^{[iW]}(t) = K_{Ha}^{[iW]} \left[1 - \beta_{Ha}^{[iW]} + \frac{\beta_{Ha}^{[iW]}}{D + 1/\tau_{Ha}^{[iW]}} D \right] \{ w_{Ha}^{[iW]}(t) \}^{p_{Ha}^{[iW]}} \quad (3a)$$

$$f_{Hb}(t) = K_{Hb} \left[1 - \sum_{i_{Hb}=1}^{I_{Hb}} \beta_{Hb}^{[i_{Hb}]} + \sum_{i_{Hb}=1}^{I_{Hb}} \frac{\beta_{Hb}^{[i_{Hb}]}}{D + 1/\tau_{Hb}^{[i_{Hb}]}} D \right] \{ w_{Hb}(t) \}^{p_{Hb}} \quad (3b)$$

where $D (= d/dt)$ is a differential operator, $w_{Ha}^{[iW]}(t)$ and $w_{Hb}(t)$ are the compression of the hammer felt, $K_{Ha}^{[iW]}$ and K_{Hb} are the stiffness coefficients, $p_{Ha}^{[iW]}$ and p_{Hb} are the positive real numbers representing nonlinearity, $\tau_{Ha}^{[iW]}$ and $\tau_{Hb}^{[i_{Hb}]}$ are the relaxation times, and $\beta_{Ha}^{[iW]}$ and $\beta_{Hb}^{[i_{Hb}]}$ are the positive real numbers with $\beta_{Ha}^{[iW]} < 1$ and $\sum_{i_{Hb}=1}^{I_{Hb}} \beta_{Hb}^{[i_{Hb}]} < 1$. The subscripts Ha and Hb denote the upper and lower layers of the hammer felt, respectively. i_{Hb} and I_{Hb} are the index and number of the spring-dashpot element (i.e., Maxwell model) in the lower layer, respectively.

When the hammer tip contacts the string surface, the integral type of the force-compression relational expression of the upper layer is derived from Eq. (3a) as follows, incorporating the displacement of the string:

$$f_{Ha}^{[iW]}(t) = K_{Ha}^{[iW]} \left[\{ w_{Ha}^{[iW]}(t) \}^{p_{Ha}^{[iW]}} + \int_0^t \{ w_{Ha}^{[iW]}(t - \tau) \}^{p_{Ha}^{[iW]}} \left\{ -\frac{\beta_{Ha}^{[iW]}}{\tau_{Ha}^{[iW]}} \exp\left(-\frac{\tau}{\tau_{Ha}^{[iW]}}\right) \right\} d\tau \right] \quad (4)$$

$$\text{with } f_{Ha}^{[iW]}(t) \geq 0, \quad w_{Ha}^{[iW]}(t) = u_{Hab}(t) - \sum_{s=z,x,y} u_s(x_H^{[iW]}, t) e_{Hs} \geq 0$$

where $u_{Hab}(t)$ is the displacement of the joint part of the upper and lower layers, $u_s(x, t)$ the s -direction component of the string displacement ($s = z, x, y$), and $x_H^{[iW]}$ the x coordinate of the striking point. $e_{Hz} = \cos \varphi_1 \cos \varphi_2$, $e_{Hx} = \sin \varphi_1$, $e_{Hy} = \cos \varphi_1 \sin \varphi_2$, where φ_1 is the angle obtained by subtracting “the angle of the hammer-motion direction from the x -axis” from the right angle, and φ_2 is the angle that “the plane, including the hammer-motion direction and the x -axis,” makes with the z -axis. The recovery amount of the felt compression of the upper layer after the hammer tip leaves the string is obtained by solving the linear homogeneous ordinary differential equation with respect to $\{ w_{Ha}^{[iW]}(t) \}^{p_{Ha}^{[iW]}}$, which is obtained by replacing $f_{Ha}^{[iW]}(t)$ in Eq. (3a) with 0.

Similarly, when at least one hammer tip contacts the string of a certain key, i.e., when at least one string of the certain key contacts the hammer, the integral type of the force-compression relational expressions of the lower layer is derived from Eq. (3b)

击锤力-压缩关系

式 (3a)、(3b) 分别给出上层和下层击锤毡的非线性广义 Maxwell 力-压缩关系；公式保持原式，见左页。D=d/dt 为微分算子。w_Ha^[iW](t) 与 w_Hb(t) 是上层/下层压缩量；K_Ha^[iW]、K_Hb 为刚度系数；p_Ha^[iW]、p_Hb 为非线性指数；τ 为松弛时间；β 为各 Maxwell 支路的权重。上层与下层分别以 Ha、Hb 标记；iHb 与 I_Hb 表示下层 Maxwell 弹簧-阻尼支路的索引与数量。

当锤尖与弦表面接触时，由式 (3a) 推得积分形式式 (4)。该式显式包含弦位移。接触时的压缩量等于上下层连接点位移 u_Hab 与击弦点处弦三维位移沿击锤运动方向投影的差。方向余弦 e_Hz、e_Hx、e_Hy 由两个几何角 φ1、φ2 决定。

锤尖离弦后，上层毡恢复通过把式 (3a) 中接触力置零后得到的线性齐次常微分方程求解。

as follows, incorporating the displacement of the hammer mass:

$$f_{Hb}(t) = K_{Hb} \left[\{w_{Hb}(t)\}^{p_{Hb}} + \int_0^t \{w_{Hb}(t-\tau)\}^{p_{Hb}} \sum_{i_{Hb}=1}^{I_{Hb}} \left\{ -\frac{\beta_{Hb}^{[i_{Hb}]}}{\tau_{Hb}^{[i_{Hb}]}} \exp\left(-\frac{\tau}{\tau_{Hb}^{[i_{Hb}]}}\right) \right\} d\tau \right]$$

with $f_{Hb}(t) \geq 0, \quad w_{Hb}(t) = u_H(t) - u_{Hab}(t) \geq 0$

(5)

The recovery amount of felt compression of the lower layer after all the hammer tips leave the strings of a certain key is obtained by solving the linear homogeneous ordinary differential equation with respect to $\{w_{Hb}(t)\}^{p_{Hb}}$, which is obtained by replacing $f_{Hb}(t)$ in Eq. (3b) with 0.

2.2 Strings

This subsection formulates a three-dimensional geometrically nonlinear string model with anisotropic end supports, enabling accurate reproduction of string polarization exchange and ringing phenomena.

In the range below nearly the 40th key (the C4 note) of a standard 88-key piano, sounds like a bell ringing or sharp, metallic sounds are frequently heard. These distinct nonharmonic sounds, which are often referred to as the “ringing sound”, become more prominent when the string is struck hard. If the level of the ringing sounds is too high, then it can be perceived as unpleasant, whereas if there are no such sounds at all, then the piano sound would be monotonous and uninteresting. Therefore, it is desirable to design the ringing sounds to be moderately audible. The analysis of the ringing sounds of a piano led to the discovery of the following phenomena: (a) In piano partial series, there is a subseries whose frequency deviations from the harmonic series are nearly 1/4 of that of the main series [9]. (b) When “the frequency of the odd-order partials of the subseries” matches “the odd-order natural frequency of the longitudinal vibration of the string,” or when “the frequency of the even-order partials of the subseries” matches “the even-order natural frequency of the longitudinal vibration of the string,” the level of the partials of the subseries, i.e., the ringing sound, increases. More analytically, (b’) When “the sum of the odd-order and the even-order natural frequencies of the transverse vibration of the string” matches “the odd-order natural frequency of the longitudinal vibration of the string,” or when “the sum of a set of the odd-order natural frequencies” or “the sum of a set of the even-order natural frequencies of the transverse vibration of the string” matches “the even-order natural frequency of the longitudinal vibration of the string,” the ringing sound increases [10].

These phenomena, which have been experimentally confirmed, can be explained using the string model, which incorporates stiffness and geometrical nonlinearity under the simplification that “the coupling between the transverse vibration and longitudinal vibration is unidirectional [4].” However, models with such simplification do not satisfy the energy conservation law based on Newton’s laws. Therefore, to obtain an appropriate solution that satisfies this law, a nonlinear string model incorporating the bidirectional coupling should be employed [5]. Furthermore, to quantitatively simulate piano sounds, including the ringing sound, with higher precision, it is crucial to

击锤模型（续）

只要某个琴键对应的至少一根弦仍与击锤接触，下层击锤毡的力-压缩关系可写成式（5）的积分形式，并显式包含击锤质量块位移。式（5）中的积分核由多个下层 Maxwell 支路的 β 与松弛时间 τ 共同决定。

所有锤尖都离开该键的弦后，下层毡的恢复同样通过把式（3b）中 f_{Hb} 置零后得到的齐次常微分方程求得。

2.2 琴弦

本节建立具有各向异性端支承的三维几何非线性琴弦模型，以准确重现偏振能量交换与 ringing。标准 88 键钢琴大约 C4 以下经常能听到钟鸣般或尖锐金属的非谐声音，强击时更显著。过多会刺耳，完全没有则声音单调，因此需要适度设计。

实验发现：钢琴分音序列中存在一个子序列，其相对谐波序列的频率偏移约为主序列的 1/4；当该子序列的特定奇/偶阶分音与纵向振动相应奇/偶阶固有频率匹配时，ringing 增强。更一般地，当若干横向固有频率之和与纵向固有频率满足特定匹配关系时，ringing 也增强。过去的单向“横向→纵向”模型可解释这些现象，但不满足完整能量守恒，因此本文采用纵向与横向的双向耦合。

model the mechanism of 3D coupling between the spatial string and the string support ends on the body (bridge and bearing), in addition to the elements considered in previous studies. Herein, a spatial string with losses, stiffness, and geometrical nonlinearity and the 3D coupling between the string and the string support ends on the body–air system are precisely described.

Considering the initial strain and geometrical nonlinearity of the Timoshenko beam model, the strain–displacement relationship of a string can be expressed as follows:

$$\begin{aligned}\varepsilon_x(x,t) &= \varepsilon_0 + \left\{ \left(1 + \frac{\partial}{\partial x}u_x\right)^2 + \left(\frac{\partial}{\partial x}u_y\right)^2 + \left(\frac{\partial}{\partial x}u_z\right)^2 \right\}^{\frac{1}{2}} - z \frac{\partial}{\partial x}\psi_z(x,t) - y \frac{\partial}{\partial x}\psi_y(x,t) - 1 \\ \gamma_{xz}(x,t) &= -\psi_z(x,t) + \frac{\partial}{\partial x}u_z(x,t), \quad \gamma_{xy}(x,t) = -\psi_y(x,t) + \frac{\partial}{\partial x}u_y(x,t)\end{aligned}$$

where $\varepsilon_x(x,t)$ is the normal strain of the string, ε_0 is the initial normal strain, $\gamma_{xz}(x,t)$ and $\gamma_{xy}(x,t)$ are the shear strain of the string, and $\psi_s(x,t)$ ($s = z, y$) are the rotation angle of the cross-section with respect to the s –axis. Furthermore, considering the linear stress–strain relationship, viscous damping, and external forces, the motion of “the speaking part of the spatial string” is expressed with five variables (three translations and two rotations) as follows:

Translation in the z – and y –axes

$$\begin{aligned}& \left\{ c'_0 \left(\frac{\partial^2}{\partial t^2} + b'_{zy} \frac{\partial}{\partial t} \right) - c'_7 \frac{\partial^2}{\partial x^2} \right\} u_s(x,t) + c'_6 \frac{\partial}{\partial x} \psi_s(x,t) + g_s^{nonl}(x,t) \\ &= f_{Ha}(t) e_{Hs} \delta(x - x_H) - \sum_{i_D=1}^{I_D} f_{Ds}^{[i_D]}(t) \delta(x - x_D^{[i_D]}) \qquad s = z, y\end{aligned}\tag{6a}$$

$$\text{with } g_s^{nonl}(x,t) = -c'_3 \frac{\partial}{\partial x} \left[\frac{\partial}{\partial x} u_s \left[1 - \left\{ \left(1 + \frac{\partial}{\partial x}u_x\right)^2 + \left(\frac{\partial}{\partial x}u_y\right)^2 + \left(\frac{\partial}{\partial x}u_z\right)^2 \right\}^{-\frac{1}{2}} \right] \right]$$

Rotation in the zx – and yx –planes

$$c'_6 \frac{\partial}{\partial x} u_s(x,t) + \left\{ -c'_5 \left(\frac{\partial^2}{\partial t^2} + b'_{zy} \frac{\partial}{\partial t} \right) - c'_6 + c'_4 \frac{\partial^2}{\partial x^2} \right\} \psi_s(x,t) = 0 \qquad s = z, y\tag{6b}$$

Translation in the x –axis

$$\begin{aligned}& \left\{ c'_0 \left(\frac{\partial^2}{\partial t^2} + b'_x \frac{\partial}{\partial t} \right) - c'_2 \frac{\partial^2}{\partial x^2} \right\} u_x(x,t) + g_x^{nonl}(x,t) \\ &= f_{Ha}(t) e_{Hx} \delta(x - x_H) - \sum_{i_D=1}^{I_D} f_{Dx}^{[i_D]}(t) \delta(x - x_D^{[i_D]})\end{aligned}\tag{6c}$$

三维几何非线性琴弦

为更精确地定量模拟钢琴声，除了既有因素外，还必须描述空间琴弦与琴桥/弦枕等琴体支承端之间的三维耦合。

本文精确描述具有损耗、刚度、几何非线性的空间弦，以及弦与琴体–空气系统支承端之间的三维连接。

考虑初始应变与 Timoshenko 梁几何非线性后，式（6）前给出弦的应变–位移关系：轴向正应变 ε_x 同时依赖初始应变 ε_0 、轴向位移梯度和两个横向位移梯度，并包含截面转动项；两个剪切应变 γ_{xz} 、 γ_{xy} 等于横向位移梯度与相应截面转角之差。

有效弦段用五个场变量描述：三个平移 u_x 、 u_y 、 u_z 与两个截面转角 ψ_y 、 ψ_z 。式（6a）给出 z 、 y 两个横向方向的平移动力学；式（6b）给出两个弯曲平面中的截面转动；式（6c）给出 x 方向纵向平移。击锤力以 δ 在击弦点进入，制音器力以 δ 在各接触点进入。几何非线性项 g_s^{nonl} 将三个方向的位移梯度耦合在一起。

with $g_x^{nonl}(x,t) = c'_3 \frac{\partial}{\partial x} \left[\left(1 + \frac{\partial}{\partial x} u_x \right) \left\{ \left(1 + \frac{\partial}{\partial x} u_x \right)^2 + \left(\frac{\partial}{\partial x} u_y \right)^2 + \left(\frac{\partial}{\partial x} u_z \right)^2 \right\}^{-\frac{1}{2}} \right]$

where $g_s^{nonl}(x,t)$ ($s = z, x, y$) are the terms caused by the geometrical nonlinearity. In pianos, wrapped strings are used for the low range, whereas plain strings are used for the mid–high range. The former consists of the core and copper wires, and the latter consists of only the core wire. c'_0 ($= \rho_f S$ for plain strings) is the linear density, c'_1 ($= ES\varepsilon_0$) the tension, c'_2 ($= ES$) the extensional rigidity, $c'_3 = c'_2(1 - \varepsilon_0)$. c'_4 ($= EI$) is the flexural rigidity, c'_5 ($= \rho_f I$ for plain strings) the rotational inertia, and c'_6 ($= kGS$) the shear rigidity. $c'_7 = c'_1 + c'_6$. ρ_f is the density of the core wire, E and G are the moduli of the longitudinal and transverse elasticity of the core wire, respectively, and k is the nondimensional Timoshenko shear coefficient. b'_{zy} and b'_x are the transverse and longitudinal mass proportional viscous damping coefficients of the string, respectively, S ($= \pi(d_f)^2/4$) the cross-sectional area, I ($= \pi(d_f)^4/64$) the second moment of area, and d_f the diameter of the core wire. For the wrapped strings, by approximating the helical structure of the copper wire with multiple donut-shaped structures, c'_0 and c'_5 are calculated using ρ_f , ρ_c , d_f , and d_c , where ρ_c and d_c are the density and diameter of the copper wire, respectively. Here, the small increase in stiffness due to the core wire is neglected. $f_{Ds}^{[ip]}(t)$ ($s = z, x, y$) are the forces exerted by the damper on the string, $x_D^{[ip]}$ are the damping points, i_D and I_D are the index and number of the damper tip in contact with the certain string, respectively, and δ is the Dirac delta function. The equations of motion of “the rear duplex part of the spatial string” are expressed by removing the nonlinear terms $g_s^{nonl}(x,t)$ ($s = z, x, y$) and the external force terms $f_{Ha}(t)$, $f_{Ds}^{[ip]}(t)$ ($s = z, x, y$) from Eq. (6).

Fig. 4 shows an image of a bridge, which is one of the string support parts. Two bridge pins are embedded apart in the bridge in the direction of the string, forming an angle with the vertical plane in opposite directions. This string end-support condition is often referred to as the zig-zag end condition (see Chaigne et al. [11], Tan [12]). The boundary condition of the string modeled based on the beam theory must be defined on its translation and rotation. The translation of the string at its support ends is constrained by the bridge and the cast-iron frame, and the rotation of the string at its support ends can be considered to be somewhere between “constrained” and “unconstrained.” Such intermediate conditions can be modeled using the rotational spring, as shown in Fig. 5. Considering the end of the bridge side, the string is constrained more firmly by the relatively stiffer metal bridge pin than by the relatively less stiff wooden bridge top. On the other hand, considering the end of the cast-iron frame side, the string is constrained more firmly in the normal direction than in the tangent direction to the cast-iron frame. In addition, the tilt of the bridge pin yields a clear anisotropy between the two support ends, as shown in Fig. 6.

琴弦物理参数与后弦段

式 (6) 中 g_s^{nonl} 是几何非线性产生的项。低音使用缠弦，中高音使用裸弦。裸弦参数: $c'_0 = \rho_f S$ 为线密度; $c'_1 = ES\varepsilon_0$ 为张力; $c'_2 = ES$ 为轴向刚度; $c'_3 = c'_2(1 - \varepsilon_0)$; $c'_4 = EI$ 为弯曲刚度; $c'_5 = \rho_f I$ 为转动惯量; $c'_6 = kGS$ 为剪切刚度; $c'_7 = c'_1 + c'_6$ 。 ρ_f 是芯弦密度, E/G 为纵向/横向弹性模量, k 为无量纲 Timoshenko 剪切系数, S 为截面积, I 为截面二次矩。

缠弦把铜绕组的螺旋结构近似为多个环形结构, 用 ρ_f 、 ρ_c 、 d_f 、 d_c 等计算线密度与转动惯量; 忽略绕线对芯弦刚度造成的小幅增加。

制音器在多个接触点施力。后弦段的控制方程由式 (6) 去除几何非线性项以及击锤/制音器外力项得到。

图 4-6 说明琴桥 zig-zag 端部条件: 两枚桥钉沿弦方向错开并向相反方向倾斜。基于梁理论, 边界既要约束平移也要处理转动; 转动介于完全固定与完全自由之间, 本文用转动弹簧描述。金属桥钉、木质桥顶及铸铁板法向/切向刚度不同, 加上桥钉倾角, 导致端部明确的各向异性。



Fig. 4 Zig-zag end condition at the bridge.



Fig. 5 Rotational-spring support model at string ends. K_B denotes the rotational spring constant.

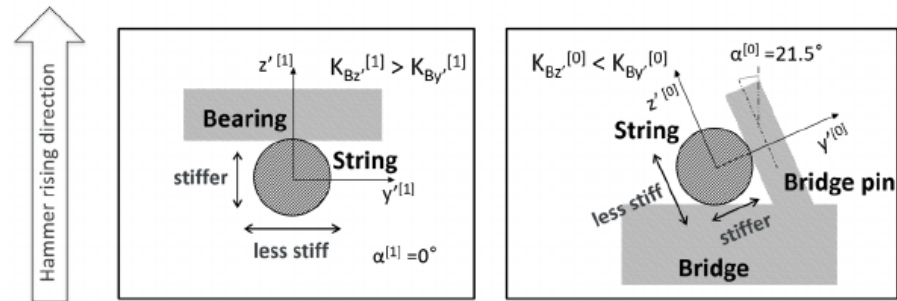


Fig. 6 Cross-sectional view of anisotropic string end support.

图 4-6

图 4: 琴桥处的 zig-zag 端部条件。两枚桥钉沿弦轴错开并以相反方向倾斜，使弦在桥上形成折线式约束。

图 5: 弦端转动弹簧支承模型。 K_B 表示转动弹簧常数，用来描述端部转动既非完全固定、也非完全自由的中间状态。

图 6: 各向异性弦端支承横截面。琴桥侧因桥钉倾角、金属桥钉与木质桥顶刚度差，两个横向方向刚度不同；铸铁板/弦枕侧法向与切向刚度也不同。

这些结构性的各向异性是后文“双偏振耦合”能够自然出现的关键，而不是人为加入一个经验耦合项。

Based on the above considerations, the boundary conditions of “the speaking part of the spatial string” are expressed as follows:

Translation in the z -, x - and y -axes

$$u_s(x, t) = u_{Bs}^{[i_B]}(t) \quad x = (-1)^{i_B} l^{[1]}; \ i_B = 0, 1; \ s = z, x, y \quad (7a)$$

Rotation in the $z'^{[i_B]} x$ -plane

$$\begin{aligned} & \cos \alpha^{[i_B]} \left[c_4' \frac{\partial}{\partial x} \psi_z(x, t) + (-1)^{i_B} K_{Bz'}^{[i_B]} \left\{ \psi_z(x, t) - \theta_{Bz}^{[i_B]}(t) \right\} \right] \\ & - \sin \alpha^{[i_B]} \left[c_4' \frac{\partial}{\partial x} \psi_y(x, t) + (-1)^{i_B} K_{By'}^{[i_B]} \left\{ \psi_y(x, t) - \theta_{By}^{[i_B]}(t) \right\} \right] = 0 \\ & x = (-1)^{i_B} l^{[1]}; \ i_B = 0, 1 \end{aligned} \quad (7b)$$

Rotation in the $y'^{[i_B]} x$ -plane

$$\begin{aligned} & \sin \alpha^{[i_B]} \left[c_4' \frac{\partial}{\partial x} \psi_z(x, t) + (-1)^{i_B} K_{By'}^{[i_B]} \left\{ \psi_z(x, t) - \theta_{Bz}^{[i_B]}(t) \right\} \right] \\ & + \cos \alpha^{[i_B]} \left[c_4' \frac{\partial}{\partial x} \psi_y(x, t) + (-1)^{i_B} K_{Bz'}^{[i_B]} \left\{ \psi_y(x, t) - \theta_{By}^{[i_B]}(t) \right\} \right] = 0 \\ & x = (-1)^{i_B} l^{[1]}; \ i_B = 0, 1 \end{aligned} \quad (7c)$$

where $u_{Bs}^{[i_B]}(t)$ ($s = z, x, y$) and $\theta_{Bs}^{[i_B]}(t)$ ($s = z, y$) are the translation displacement and rotation angles at the support ends, i.e., the portions of the body, respectively. $i_B = 0$ and $i_B = 1$ denote the bridge and cast-iron frame (bearing) sides of the speaking part of the string, respectively, and $\alpha^{[0]}$ and $\alpha^{[1]}$ denote the inclination angle of the bridge pin and the normal to the cast-iron frame with respect to the zx -plane, respectively. The $xy'z'^{[i_B]}$ -coordinate system is defined as one rotated clockwise the xyz -coordinate system around the x -axis by $\alpha^{[i_B]}$. $K_{Bz'}^{[i_B]}$ and $K_{By'}^{[i_B]}$ denote the rotational spring constants around the $y'^{[i_B]}$ - and $z'^{[i_B]}$ -axes, respectively. They depend on the flexural rigidity and bending angle of the core wire. $K_{Bz'}^{[0]}$ also depends on the cut depth of the bridge top. The boundary conditions of “the rear duplex part of the spatial string” are represented by replacing $l^{[1]}$ in Eq. (7) with $l^{[2]}$. Notably, the zig-zag end condition is naturally modeled by considering the tilt of the bridge pins at both the speaking and rear duplex parts of the string.

When the hybrid-interface CMS (using the component modes calculated under the condition that the strings are unconstrained by the hammers/dampers and constrained by the body-air system) is employed to represent the displacement of the string influenced by the motions of the hammer/damper and body-air system, the translation displacements and rotation angles of “the speaking part of the spatial string” are given as expressed below. This method is usually used with FEM; however, herein, it is used with the semi-analytic method, which makes it easy to observe

弦端边界条件与混合界面 CMS

式 (7a)–(7c) 给出有效弦段边界条件。式 (7a) 规定两端在 z 、 x 、 y 三方向的平移等于琴体对应支承点的平移 $u_{Bs}^{[i_B]}$ 。 $i_B=0$ 表示琴桥侧， $i_B=1$ 表示铸铁板/弦枕侧。

式 (7b)、(7c) 分别描述绕两个旋转后的局部轴的转动边界； $\alpha^{[0]}$ 与 $\alpha^{[1]}$ 分别表示桥钉倾角和铸铁板法线相对于 zx 平面的角度。 $K_{Bz'}$ 、 $K_{By'}$ 是两个主方向的转动弹簧常数，它们与芯弦弯曲刚度、弯折角等有关；琴桥侧某一方向还受桥顶切槽深度影响。后弦段以 $l^{[2]}$ 替换 $l^{[1]}$ 得到同类边界。

随后采用混合界面 CMS：自然模态是在“弦不受击锤/制音器约束、但受琴体-空气系统约束”的条件下计算。弦的平移与转角由自然模态项加约束（静态变形）模态项表示。这种方法通常用于 FEM，这里与半解析弦模型结合，便于直接观察物理参数与解之间的关系。

the relationship between physical parameters and solutions.

$$u_s(x,t)=\sum_{r_{zy}=1}^{R_{zy}}\xi_s^{[r_{zy}]}(t)U_s^{[r_{zy}]}(x)+\tilde{u}_s(x,t)\quad s=z,y\tag{8a}$$

$$\psi_s(x,t)=\sum_{r_{zy}=1}^{R_{zy}}\xi_s^{[r_{zy}]}(t)\Psi_s^{[r_{zy}]}(x)+\tilde{\psi}_s(x,t)\quad s=z,y\tag{8b}$$

$$u_x(x,t)=\sum_{r_x=1}^{R_x}\xi_x^{[r_x]}(t)U_x^{[r_x]}(x)+\tilde{u}_x(x,t)\tag{8c}$$

where $U_s^{[r_{zy}]}(x)$ ($s=z,y$) and $U_x^{[r_x]}(x)$ are the hybrid-interface modes of the transverse and longitudinal translation of the string, respectively. $\Psi_s^{[r_{zy}]}(x)$ ($s=z,y$) are the hybrid-interface modes of the rotation angle of the string, $\xi_s^{[r_{zy}]}(t)$ ($s=z,y$) and $\xi_x^{[r_x]}(t)$ are the modal coordinates, r_{zy} and r_x are the indices of the natural mode, R_{zy} and R_x are the numbers of the considered mode, and $\tilde{u}_s(x,t)$ ($s=z,x,y$) and $\tilde{\psi}_s(x,t)$ ($s=z,y$) are the constrained (static deformation) modes of translation and rotation of the string, respectively.

The hybrid-interface modes (natural modes) are represented below by applying the variable separation method to the homogeneous equations, which removes the nonlinear, external force, and attenuation terms from Eq. (6).

$$\begin{aligned} U_s^{[r_{zy}]}(x) = & A_s^{[r_{zy}]} \cosh(\mu_{zy}^{[r_{zy}]}x) + B_s^{[r_{zy}]} \sinh(\mu_{zy}^{[r_{zy}]}x) \\ & + C_s^{[r_{zy}]} \cos(\nu_{zy}^{[r_{zy}]}x) + D_s^{[r_{zy}]} \sin(\nu_{zy}^{[r_{zy}]}x) \\ & r_{zy} = 1, 2, \cdots, R_{zy}; s = z, y \end{aligned}\tag{9a}$$

$$\begin{aligned} \Psi_s^{[r_{zy}]}(x) = & A_s^{[r_{zy}]} \hat{\mu}_{zy}^{[r_{zy}]} \sinh(\mu_{zy}^{[r_{zy}]}x) + B_s^{[r_{zy}]} \hat{\mu}_{zy}^{[r_{zy}]} \cosh(\mu_{zy}^{[r_{zy}]}x) \\ & - C_s^{[r_{zy}]} \hat{\nu}_{zy}^{[r_{zy}]} \sin(\nu_{zy}^{[r_{zy}]}x) + D_s^{[r_{zy}]} \hat{\nu}_{zy}^{[r_{zy}]} \cos(\nu_{zy}^{[r_{zy}]}x) \\ & r_{zy} = 1, 2, \cdots, R_{zy}; s = z, y \end{aligned}\tag{9b}$$

$$U_x^{[r_x]}(x) = C_x^{[r_x]} \cos(\nu_x^{[r_x]}x) + D_x^{[r_x]} \sin(\nu_x^{[r_x]}x) \quad r_x = 1, 2, \cdots, R_x \tag{9c}$$

with

$$\begin{aligned} \mu_{zy}^{[r_{zy}]} = & \left[\{(\alpha_{zy}^{[r_{zy}]})^2 + \beta_{zy}^{[r_{zy}]}\}^{\frac{1}{2}} + \alpha_{zy}^{[r_{zy}]}\right]^{\frac{1}{2}}, \quad \nu_{zy}^{[r_{zy}]} = \left[\{(\alpha_{zy}^{[r_{zy}]})^2 + \beta_{zy}^{[r_{zy}]}\}^{\frac{1}{2}} - \alpha_{zy}^{[r_{zy}]}\right]^{\frac{1}{2}} \\ \alpha_{zy}^{[r_{zy}]} = & -\frac{1}{2} \left\{ \frac{c'_0(\Omega_{zy}^{[r_{zy}]})^2}{c'_7} + \frac{c'_5(\Omega_{zy}^{[r_{zy}]})^2 - c'_6}{c'_4} + \frac{(c'_6)^2}{c'_4 c'_7} \right\}, \quad \beta_{zy}^{[r_{zy}]} = \frac{c'_0(\Omega_{zy}^{[r_{zy}]})^2 \{c'_6 - c'_5(\Omega_{zy}^{[r_{zy}]})^2\}}{c'_4 c'_7} \end{aligned}$$

弦的模态展开

式 (8a)–(8c) 将横向位移 u_s 、截面转角 ψ_s 和纵向位移 u_x 表示为：若干混合界面自然模态的线性组合 + 约束（静态变形）模态。 ξ 是对应模态坐标； R_{zy} 、 R_x 是保留的横向/纵向模态数量。

自然模态由式 (6) 去掉非线性、外力和衰减项后，通过变量分离得到。式 (9a)、(9b) 给出 Timoshenko 横向位移/转角模态，包含 $\cosh$ 、 $\sinh$ 、 $\cos$ 、 $\sin$ ；式 (9c) 给出纵向模态。 μ 、 ν 及其带帽版本由固有角频率和弦的质量、弯曲、剪切参数决定。

本页公式完整保留在左侧原页。中文侧不重排公式，以避免任何符号、上下标或括号在转写时失真。

$$\begin{aligned}\dot{\mu}_{zy}^{[r_{zy}]} &= \frac{1}{c'_6} \left\{ c'_7 \mu_{zy}^{[r_{zy}]} + c'_0 \frac{(\Omega_{zy}^{[r_{zy}]})^2}{\mu_{zy}^{[r_{zy}]}} \right\}, \quad \dot{\nu}_{zy}^{[r_{zy}]} = \frac{1}{c'_6} \left\{ c'_7 \nu_{zy}^{[r_{zy}]} - c'_0 \frac{(\Omega_{zy}^{[r_{zy}]})^2}{\nu_{zy}^{[r_{zy}]}} \right\} \\ \nu_x^{[r_x]} &= \Omega_x^{[r_x]} (c'_2/c'_0)^{-\frac{1}{2}}\end{aligned}$$

where $\Omega_{zy}^{[r_{zy}]}$ and $\Omega_x^{[r_x]}$ are the natural angular frequencies of the transverse and longitudinal vibrations, respectively. When the fundamental natural angular frequency of the transverse vibration $\Omega_{zy}^{[1]}$ is given, ε_0 and $\Omega_{zy}^{[r_{zy}]} (r_{zy} = 2, 3, \dots, R_{zy})$ are uniquely determined by numerically solving the characteristic equations, i.e., a system of transcendental equations, with $u_{Bs}^{[i_B]}(t) (i_B = 0, 1; s = z, y)$ and $\theta_{Bs}^{[i_B]}(t) (i_B = 0, 1; s = z, y)$ replaced by 0 in Eq. (7). In addition, $\Omega_x^{[r_x]} (= r_x \pi / (2l^{[1]})) (c'_2/c'_0)^{\frac{1}{2}}; r_x = 1, 2, \dots, R_x)$ are determined under the condition that $u_{Bx}^{[i_B]}(t) (i_B = 0, 1)$ is replaced with 0 in Eq. (7a). Furthermore, when normalizing the modal mass to unity, $A_s^{[r_{zy}]}, B_s^{[r_{zy}]}, C_s^{[r_{zy}]}, D_s^{[r_{zy}]}$ ($r_{zy} = 1, 2, \dots, R_{zy}; s = z, y$), $C_x^{[r_x]}$, and $D_x^{[r_x]}$ ($r_x = 1, 2, \dots, R_x$) are uniquely determined. Applying the computational method that Kondo [13] used to prove the orthogonality of natural modes, the modal mass and stiffness can be expressed as follows:

$$M_{zy}^{[r_{zy}]} = \int_{-l^{[1]}}^{l^{[1]}} \left[c'_0 \left\{ (U_z^{[r_{zy}]})^2 + (U_y^{[r_{zy}]})^2 \right\} + c'_5 \left\{ (\Psi_z^{[r_{zy}]})^2 + (\Psi_y^{[r_{zy}]})^2 \right\} \right] dx \quad r_{zy} = 1, 2, \dots, R_{zy} \quad (10a)$$

$$\begin{aligned}K_{zy}^{[r_{zy}]} &= \int_{-l^{[1]}}^{l^{[1]}} \left[c'_1 \left\{ \left(\frac{d}{dx} U_z^{[r_{zy}]} \right)^2 + \left(\frac{d}{dx} U_y^{[r_{zy}]} \right)^2 \right\} + c'_4 \left\{ \left(\frac{d}{dx} \Psi_z^{[r_{zy}]} \right)^2 + \left(\frac{d}{dx} \Psi_y^{[r_{zy}]} \right)^2 \right\} \right. \\ &\quad \left. + c'_6 \left\{ \left(-\Psi_z^{[r_{zy}]} + \frac{d}{dx} U_z^{[r_{zy}]} \right)^2 + \left(-\Psi_y^{[r_{zy}]} + \frac{d}{dx} U_y^{[r_{zy}]} \right)^2 \right\} \right] dx \\ &\quad + \sum_{i_B=0}^1 \left\{ \left(K_{Bz'}^{[i_B]} \cos^2 \alpha^{[i_B]} + K_{By'}^{[i_B]} \sin^2 \alpha^{[i_B]} \right) (\Psi_z^{[r_{zy}]})^2 \right. \\ &\quad \left. - 2 \left(K_{Bz'}^{[i_B]} - K_{By'}^{[i_B]} \right) \cos \alpha^{[i_B]} \sin \alpha^{[i_B]} \Psi_z^{[r_{zy}]} \Psi_y^{[r_{zy}]} \right. \\ &\quad \left. + \left(K_{Bz'}^{[i_B]} \sin^2 \alpha^{[i_B]} + K_{By'}^{[i_B]} \cos^2 \alpha^{[i_B]} \right) (\Psi_y^{[r_{zy}]})^2 \right\} \Big|_{x=(-1)^{i_B} l^{[1]}} \\ &\quad r_{zy} = 1, 2, \dots, R_{zy} \quad (10b)\end{aligned}$$

$$M_x^{[r_x]} = \int_{-l^{[1]}}^{l^{[1]}} c'_0 (U_x^{[r_x]})^2 dx \quad r_x = 1, 2, \dots, R_x \quad (10c)$$

$$K_x^{[r_x]} = \int_{-l^{[1]}}^{l^{[1]}} c'_2 \left(\frac{d}{dx} U_x^{[r_x]} \right)^2 dx \quad r_x = 1, 2, \dots, R_x \quad (10d)$$

固有频率、模态归一化与模态质量/刚度

$\Omega_{zy}^{[r_{zy}]}$ 和 $\Omega_x^{[r_x]}$ 分别是横向和纵向固有角频率。给定横向基频 $\Omega_{zy}^{[1]}$ 后，将支承端的平移/转角置零，数值求解由边界条件产生的超越特征方程组，即可唯一确定初始应变 ε_0 与其余横向固有频率。纵向固有频率由对应固定端条件直接确定。

把模态质量归一化为 1 后，式 (9) 中各模态系数即可唯一确定。作者采用 Kondo 用于证明自然模态正交性的计算方法，得到式 (10a)–(10d) 的横向/纵向模态质量与刚度积分。横向模态质量包含平移质量与截面转动惯量；横向模态刚度同时包含张力、弯曲、剪切能量以及两端转动弹簧储能。

where $M_{zy}^{[r_{zy}]}$ and $K_{zy}^{[r_{zy}]}(r_{zy} = 1, 2, \cdots, R_{zy})$ are the modal mass and stiffness of the transverse mode, and $M_x^{[r_x]}$ and $K_x^{[r_x]}(r_x = 1, 2, \cdots, R_x)$ are the modal mass and stiffness of the longitudinal mode, respectively. Rayleigh quotient $(\Omega_{zy}^{[r_{zy}]})^2 = K_{zy}^{[r_{zy}]} / M_{zy}^{[r_{zy}]} = K_{zy}^{[r_{zy}]}$, $(\Omega_x^{[r_x]})^2 = K_x^{[r_x]} / M_x^{[r_x]} = K_x^{[r_x]}$ can be used for consistency checks. Eq. (9b) shows that two polarities of the transverse mode can be separated (i.e., separating $M_{zy}^{[r_{zy}]}$ into $M_z^{[r_z]}$ and $M_y^{[r_y]}$; separating $K_{zy}^{[r_{zy}]}$ into $K_z^{[r_z]}$ and $K_y^{[r_y]}$) only when $(K_{Bz'}^{[i_B]} - K_{By'}^{[i_B]}) \cos \alpha^{[i_B]} \sin \alpha^{[i_B]} = 0 (i_B = 0, 1)$.

The constrained modes are represented by solving the homogeneous equations, removing the nonlinear, external force, and time derivative terms from Eq. (6), as follows:

$$\tilde{u}_s(x, t) = a_s(t) + b_s(t) x + c_s(t) \cosh(c_{14}c_{67}x) + d_s(t) \sinh(c_{14}c_{67}x) \quad s = z, y \tag{11a}$$

$$\tilde{\psi}_s(x, t) = b_s(t) + c_s(t) \frac{c_{14}}{c_{67}} \sinh(c_{14}c_{67}x) + d_s(t) \frac{c_{14}}{c_{67}} \cosh(c_{14}c_{67}x) \quad s = z, y \tag{11b}$$

$$\begin{aligned} \tilde{u}_x(x, t) &= a_x(t) + b_x(t) x \\ \text{with } c_{14} &= (c'_1/c'_4)^{\frac{1}{2}}, c_{67} = (c'_6/c'_7)^{\frac{1}{2}} \end{aligned} \tag{11c}$$

If $u_{Bs}^{[i_B]}(t) (i_B = 0, 1; s = z, x, y)$ and $\theta_{Bs}^{[i_B]}(t) (i_B = 0, 1; s = z, y)$ are given (see Eq. (32)), $a_s(t)$, $b_s(t)$, $c_s(t)$, $d_s(t) (s = z, y)$, $a_x(t)$, and $b_x(t)$ are uniquely determined under the boundary condition (Eq. (7)).

To integrate the nonlinear partial differential equations (Eq. (6)) with respect to variable x, the Galerkin method [14] is employed. This involves substituting Eq. (8) into Eqs. (6a) and (6c), multiplying both sides by $U_s^{[r_{zy}]}(x) (r_{zy} = 1, 2, \cdots, R_{zy}; s = z, y)$ and $U_x^{[r_x]}(x) (r_x = 1, 2, \cdots, R_x)$, respectively, and integrating them from $x = -l^{[1]}$ to $x = l^{[1]}$ with respect to variable x , and the following simultaneous nonlinear ordinary differential equations are obtained.

$$\begin{aligned} &\left\{ \frac{d^2}{dt^2} + b'_{zy} \frac{d}{dt} + (\Omega_{zy}^{[r_{zy}]})^2 \right\} \xi_{zy}^{[r_{zy}]}(t) + \int_{-l^{[1]}}^{l^{[1]}} g_s^{nonl}(x, t) U_s^{[r_{zy}]}(x) dx \\ &+ c'_0 \left(\frac{\partial^2}{\partial t^2} + b'_{zy} \frac{\partial}{\partial t} \right) \int_{-l^{[1]}}^{l^{[1]}} \tilde{u}_s(x, t) U_s^{[r_{zy}]}(x) dx \\ &= f_{Ha}(t) e_{Hs} U_s^{[r_{zy}]}(x_H) - \sum_{i_D=1}^{I_D} f_{Ds}^{[i_D]}(t) U_s^{[r_{zy}]}(x_D^{[i_D]}) \quad r_{zy} = 1, 2, \cdots, R_{zy}; s = z, y \end{aligned} \tag{12a}$$

双偏振的不可分离性、约束模态与 Galerkin 离散

$M_{zy}^{[r_{zy}]}$ 、 $K_{zy}^{[r_{zy}]}$ 是横向模态质量/刚度, $M_x^{[r_x]}$ 、 $K_x^{[r_x]}$ 是纵向模态质量/刚度。Rayleigh 商 $\Omega^2 = K/M$ 可用于一致性检查。

一个关键结论是：只有当端部两个主方向的转动刚度差与倾角组合项为零时，两个横向偏振才能完全分离。一般真实钢琴并不满足这一条件，因此端部各向异性会自然导致两种横向偏振耦合。

约束模态由式（6）去除非线性、外力和时间导数后求得，见式（11）。给定琴体支承端的平移与转角后，各约束模态系数唯一确定。

为对非线性 PDE 的空间变量积分，作者采用 Galerkin 法：将式（8）的模态展开代入式（6a）、（6c），分别乘以对应自然模态并沿整根有效弦长积分，得到一组同时耦合的非线性常微分方程（式 12）。

$$\begin{aligned} & \left\{ \frac{d^2}{dt^2} + b'_x \frac{d}{dt} + (\Omega_x^{[r_x]})^2 \right\} \xi_x^{[r_x]}(t) + \int_{-l^{[1]}}^{l^{[1]}} g_x^{nonl}(x, t) U_x^{[r_x]}(x) dx \\ & + c'_0 \left(\frac{\partial^2}{\partial t^2} + b'_x \frac{\partial}{\partial t} \right) \int_{-l^{[1]}}^{l^{[1]}} \tilde{u}_x(x, t) U_x^{[r_x]}(x) dx \\ & = f_{Ha}(t) e_{Hx} U_x^{[r_x]}(x_H) - \sum_{i_D=1}^{I_D} f_{Dx}^{[i_D]}(t) U_x^{[r_x]}(x_D^{[i_D]}) \quad r_x = 1, 2, \dots, R_x \quad (12b) \end{aligned}$$

To describe the frequency dependence of string losses in detail, b'_{zy} and b'_x are replaced by $b'_{zy}{}^{[r_{zy}]}(r_{zy} = 1, 2, \dots, R_{zy})$ and $b'_x{}^{[r_x]}(r_x = 1, 2, \dots, R_x)$, which are expressed as follows:

$$\begin{aligned} b'_{zy}{}^{[r_{zy}]} &= 2 \zeta_{zy}^{[r_{zy}]} \omega \simeq 2 \zeta_{zy}^{[r_{zy}]} \Omega_{zy}^{[r_{zy}]} \\ \zeta_{zy}^{[r_{zy}]} &= \left\{ \alpha' \omega^{-1} + \nu' \omega^{-\frac{1}{2}} + \eta'_{zy} \omega^0 + \beta' \omega^{p'} \right\} / 2 \\ &\simeq \left\{ \alpha' (\Omega_{zy}^{[r_{zy}]})^{-1} + \nu' (\Omega_{zy}^{[r_{zy}]})^{-\frac{1}{2}} + \eta'_{zy} (\Omega_{zy}^{[r_{zy}]})^0 + \beta' (\Omega_{zy}^{[r_{zy}]})^{p'} \right\} / 2 \\ \alpha' &= 4\pi \rho_a \eta_a / c'_0, \quad \nu' = \alpha' (2/\eta_a)^{\frac{1}{2}} r_e \\ b'_x{}^{[r_x]} &= 2 \zeta_x^{[r_x]} \omega \simeq 2 \zeta_x^{[r_x]} \Omega_x^{[r_x]} \\ \zeta_x^{[r_x]} &= \left\{ \eta'_x \omega^0 + \beta' \omega^{p'} \right\} / 2 \simeq \left\{ \eta'_x (\Omega_x^{[r_x]})^0 + \beta' (\Omega_x^{[r_x]})^{p'} \right\} / 2 \end{aligned}$$

where ω is the angular frequency. $\zeta_{zy}^{[r_{zy}]}$ and $\zeta_x^{[r_x]}$ are the transverse and longitudinal modal damping ratio of the string, respectively, and η'_{zy} and η'_x are the transverse and longitudinal structural damping coefficients of the string, respectively. Notably, η'_{zy} and η'_x include not only the internal friction of the string but also the friction between the string and support part. η'_{zy} depends on the tension of the string [15, 16]. α' and ν' are the coefficients related to the air resistance of the string [17]. β' and $p'(\geq 1)$ are the damping correction coefficient and the damping correction exponent of the string, which are introduced to compensate modal truncation errors in the body–air system. $r_e (= d_f/2$ for plain strings) is the effective radius of the string. ρ_a and η_a are the density and kinematic viscosity coefficient of air, respectively. Herein, we use an approximation in which the angular frequency is replaced by the natural angular frequency to apply the damping model originally defined in the frequency domain to the time domain. It is known that when the attenuation is small, the frequency response functions of each mode before and after approximation match well [18].

The force $f_{Bs}^{[i_B]}(t)$ ($i_B = 0, 1$; $s = z, x, y$) and the moment of force $m_{Bs}^{[i_B]}(t)$ ($i_B = 0, 1$; $s = z, y$) at the support ends are described using the translation displacement and rotation angle as follows:

$$\begin{aligned} f_{Bs}^{[i_B]}(t) &= (-1)^{i_B} \left\{ -c'_7 \frac{\partial}{\partial x} u_s(x, t) + c'_6 \psi_s(x, t) - f_s^{nonl}(x, t) \right\} \\ x &= (-1)^{i_B} l^{[1]}; \quad i_B = 0, 1; \quad s = z, y \quad (13a) \end{aligned}$$

模态时域方程与频率相关损耗

式 (12a)、(12b) 分别给出横向与纵向模态坐标的非线性 ODE。每个模态具有自身固有频率与阻尼，同时包含几何非线性项在该模态上的空间投影、由琴体支承端静态变形造成的项，以及击锤/制音器外力在模态形状上的投影。

为细致描述弦损耗的频率依赖性，原来统一的横向/纵向黏性阻尼系数被替换为逐模态的 $b_{zy}{}^{[r_{zy}]}$ 、 $b_x{}^{[r_x]}$ 。对应模态阻尼比 ξ 包含多个频率幂次项。横向结构阻尼系数不仅表示弦材料内耗，也把弦与支承部件间的摩擦并入；其值依赖张力。 α 、 ν 项描述空气阻力。 β 、 p 是用于补偿琴体–空气模态截断误差的阻尼修正。为了把原本定义在频域的阻尼模型用于时域，作者用每个模态的固有角频率近似一般角频率；在衰减较小时，近似前后各模态频响高度一致。

with $f_s^{nonl}(x,t) = c_3' \frac{\partial}{\partial x} u_s \left[1 - \left\{ \left(1 + \frac{\partial}{\partial x} u_x \right)^2 + \left(\frac{\partial}{\partial x} u_y \right)^2 + \left(\frac{\partial}{\partial x} u_z \right)^2 \right\}^{-\frac{1}{2}} \right]$

$m_{Bs}^{[i_B]}(t) = (-1)^{i_B} c_4' \frac{\partial}{\partial x} \psi_s(x,t) \qquad x = (-1)^{i_B} l^{[1]}; \ i_B = 0, 1; \ s = z, y \quad (13b)$

$f_{Bx}^{[i_B]}(t) = (-1)^{i_B} \left\{ -c_2' \frac{\partial}{\partial x} u_x(x,t) - f_x^{nonl}(x,t) \right\} \qquad x = (-1)^{i_B} l^{[1]}; \ i_B = 0, 1 \quad (13c)$

with $f_x^{nonl}(x,t) = c_3' \left[1 - \left(1 + \frac{\partial}{\partial x} u_x \right) \left\{ \left(1 + \frac{\partial}{\partial x} u_x \right)^2 + \left(\frac{\partial}{\partial x} u_y \right)^2 + \left(\frac{\partial}{\partial x} u_z \right)^2 \right\}^{-\frac{1}{2}} \right]$

Eqs. (8) to (13) are derived for the speaking part of the spatial string. The equations for the rear duplex part of the spatial string are derived similarly by removing the nonlinear and external force terms, and replacing $l^{[1]}$ with $l^{[2]}$.

2.3 Damper

In the initial state, piano strings are suppressed from vibrating by dampers. When a piano key is pressed, the corresponding damper gradually moves away from the corresponding string, and eventually the string is completely released from the resistance of the damper, preparing to be struck by the hammer. Furthermore, in a piano, the degree of contact between the damper and string can be changed not only by the depression amount of the key but also by the depression amount of the damper pedal, thereby controlling how the sound stops and the degree of string resonance. In this paper, the mechanism of the damper in such a piano is succinctly expressed using the following equation, which describes the relationship between the damper’s resistance force and the displacement of the damping point on the string.

$f_{Ds}^{[i_D]}(t) = b_{Ds} \{ 1 - e_D(t) \}^{p_D} \frac{d}{dt} u_s(x_D^{[i_D]}, t) \qquad i_D = 1, 2, \cdots, I_D; \ s = z, x, y \quad (14)$

with $b_{Ds} = K_D'' / \Omega_{zy}^{[r_{zy}]}$
 $b_{Ds} = K_D'' / \Omega_x^{[r_x]}$
 $x_D^{[i_D]} = x_H + L_D(i_D - 0.5) / I_D$
 $e_D(t) = \max(e_K(t), e_P(t))$
 $e_K(t) = 0$: The key is not depressed at all.
 $0 \leq e_K(t) \leq 1$: The key is depressed in halfway.
 $e_K(t) = 1$: The key is depressed completely.
 $e_P(t) = 0$: The damper pedal is not depressed at all.
 $0 \leq e_P(t) \leq 1$: The damper pedal is depressed in halfway.
 $e_P(t) = 1$: The damper pedal is depressed completely.

where K_D'' and p_D and the loss elastic coefficient and the nonlinear exponent of the damper, respectively. L_D is the damper length. $e_K(t)$ and $e_P(t)$ are the nondimensional depression amount of the key and damper pedal, respectively. This simple damper model means “a structural damping model with one end connected to a rigid

弦端力、制音器

式 (13a)–(13c) 用弦端的平移与转角描述琴弦在两端施加给支承结构的三维力与弯矩。横向端力同时包含线性张力/剪切项和几何非线性修正；纵向端力同样包含几何非线性项。后弦段的对应方程通过去除非线性与外力、并用 $L^{[2]}$ 替换 $L^{[1]}$ 获得。

2.3 制音器

静止时钢琴弦由制音器抑制。按键后对应制音器逐渐离弦，制音踏板也改变接触程度，从而控制止音方式与共鸣程度。本文用式 (14) 简化表示制音器：阻力与制音点处弦速度成比例，而阻尼强度由按键下压量 $e_K(t)$ 与踏板下压量 $e_P(t)$ 的较大者控制。

该模型等价于一端连刚性墙、另一端连弦上阻尼点的结构阻尼元件。作者明确指出仍有改进空间，例如没有考虑制音器质量以及弓弦式摩擦机制。

wall and the other end connected to a damping point on the string” and there is still room for improvement, such as the fact that mass and bowed string mechanism are not taken into account.

2.4 Wood as a 3D orthotropic complex elastic material

When a structure is allowed to freely vibrate in the air, the vibrations gradually attenuate over time and eventually stop. Such damping phenomena can be attributed to four factors: 1) air resistance, 2) internal friction of the material, 3) friction at the joints of each part and the support of the structure, and 4) transfer of energy to other vibration systems through the support. 2) and 3) are called structural damping. Structural damping is proportional to amplitude and independent of frequency and exhibits the same phase as velocity [19]. To describe this phenomenon, the complex stiffness constant k^* instead of real stiffness constant k is often used as follows:

$$k^* = k' + jk'' = k' (1 + j\eta)$$

where j is an imaginary unit, k' and k'' are the storage and loss stiffness constants, respectively, and $\eta (= k''/k')$ is the loss factor.

Wood is widely used for various parts of musical instruments, such as soundboards and rims. This is attributed to the lightweight and high stiffness of wood, which allows for fast wave propagation. Murase and Sawada [20] also stated that wood exhibits relatively low modal damping at low frequencies and high modal damping at high frequencies. In contrast, metals exhibit low modal damping at all frequencies, whereas plastics have high modal damping at all frequencies. This unique property of wood contributes to the gentle and pleasant sounds of wooden musical instruments. The property is attributed to the fact that when a wooden board vibrates transversely, the contribution of shear deformation increases for higher modes, and the structural damping associated with shear deformation is larger than that of elongation deformation. This property can be represented by modeling wood as a macroscopically homogeneous, 3D orthotropic, complex elastic material with the anatomical directions (L, R, T) aligned to the elastic principal axes (Fig. 7).

The stress-strain relationship in wood can be expressed as follows:

$$\begin{aligned} \{\sigma\} &= ([D'] + j[D''])\{\varepsilon\} \\ \{\sigma\} &= \{\sigma_L, \sigma_R, \sigma_T, \tau_{LR}, \tau_{RT}, \tau_{TL}\}^T, \quad \{\varepsilon\} = \{\varepsilon_L, \varepsilon_R, \varepsilon_T, \gamma_{LR}, \gamma_{RT}, \gamma_{TL}\}^T \end{aligned}$$

$$[D']^{-1} = \begin{bmatrix} 1/E'_L & -\nu_{RL}/E'_R & -\nu_{TL}/E'_T & 0 & 0 & 0 \\ -\nu_{LR}/E'_L & 1/E'_R & -\nu_{TR}/E'_T & 0 & 0 & 0 \\ -\nu_{LT}/E'_L & -\nu_{RT}/E'_R & 1/E'_T & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/G'_{RT} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/G'_{TL} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/G'_{LR} \end{bmatrix}$$

2.4 木材：三维正交各向异性复弹性材料

结构自由振动会逐渐衰减，主要来自：空气阻力；材料内部摩擦；部件连接/支承摩擦；通过支承向其他振动系统传能。其中材料内耗与连接摩擦称为结构阻尼。本文以复刚度 $k^*=k'+jk''=k'(1+j\eta)$ 描述结构阻尼，其中 k' 为储能刚度， k'' 为损耗刚度， $\eta=k''/k'$ 为损耗因子。

木材广泛用于响板 and 外壳等，因为它轻且刚度高、波传播快。已有研究指出，木材低频模态阻尼较小、高频模态阻尼较大；金属各频段阻尼都较小，塑料则普遍较大。作者把木材这种特性与高阶横向振动中剪切变形贡献增加、且剪切相关结构阻尼大于伸长变形阻尼联系起来。

因此木材被建模为宏观均匀的三维正交各向异性复弹性材料，L/R/T 解剖方向与弹性主轴一致。应力-应变关系写为 $\{\sigma\} = ([D'] + j[D''])\{\varepsilon\}$ 。

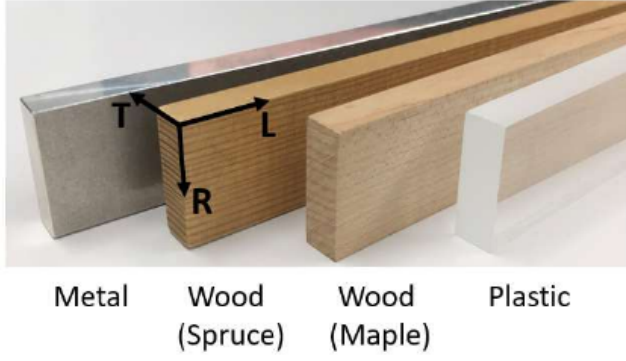


Fig. 7 Three-dimensional orthotropic material directions of wood and associated elastic constants.

$$\text{with } \nu_{RL}/E'_R = \nu_{LR}/E'_L, \quad \nu_{TL}/E'_T = \nu_{LT}/E'_L, \quad \nu_{TR}/E'_T = \nu_{RT}/E'_R$$

$$[D'']^{-1} = \begin{bmatrix} 1/E''_L & -\nu_{RL}/E''_R & -\nu_{TL}/E''_T & 0 & 0 & 0 \\ -\nu_{LR}/E''_L & 1/E''_R & -\nu_{TR}/E''_T & 0 & 0 & 0 \\ -\nu_{LT}/E''_L & -\nu_{RT}/E''_R & 1/E''_T & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/G''_{RT} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/G''_{TL} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/G''_{LR} \end{bmatrix}$$

where σ and τ are the normal and shear stress, ε and γ the normal and shear strains, $[D']$ and $[D'']$ the storage and loss elastic constant matrices, E' and E'' the longitudinal storage and loss elastic moduli, G' and G'' the transverse storage and loss elastic moduli, respectively. ν is Poisson's ratio. The loss factors are expressed as follows:

$$\eta_L = E''_L/E'_L, \quad \eta_R = E''_R/E'_R, \quad \eta_T = E''_T/E'_T, \\ \eta_{LR} = G''_{LR}/G'_{LR}, \quad \eta_{RT} = G''_{RT}/G'_{RT}, \quad \eta_{TL} = G''_{TL}/G'_{TL}$$

2.5 Manufacturing process

Without considering the nonlinearity involved in the manufacturing process, the equilibrium equation of the body can be described using FEM as follows:

$$[K'_b]\{u_b\} = \{f_b\} \quad (15)$$

where $\{u_b\} \in \mathbb{R}^{N_b}$ and $\{f\} \in \mathbb{R}^{N_b}$ are the displacement and external force vectors of the body, respectively. N_b is the number of degrees of freedom (DOF) of the finite element model of the body. It becomes large when all the parts are assembled and the manufacturing of the body is completed. $[K'_b] \in \mathbb{R}^{N_b \times N_b}$ is the stiffness matrix

木材弹性矩阵与损耗因子；2.5 制造过程

左页继续给出三维正交各向异性材料的储能弹性矩阵 $[D']$ 与损耗弹性矩阵 $[D'']$ ，并满足相应 Poisson 比互易关系。 E' 、 E'' 分别为纵向储能/损耗弹性模量， G' 、 G'' 为剪切储能/损耗模量。各主方向与剪切方向损耗因子定义为相应损耗模量与储能模量之比。

2.5 制造过程

在不考虑制造非线性情况下，琴体的 FEM 静力平衡可写为 $[K'_b]\{u_b\} = \{f_b\}$ (式 15)。其中 $\{u_b\}$ 、 $\{f_b\}$ 是琴体位移和外力向量， N_b 是有限元自由度数， $[K'_b]$ 是刚度矩阵。对于完整钢琴， N_b 非常大。

calculated based on the standard FEM theory as follows:

$$[K'_b] = \sum_m \int_V [B]^T [D'] [B] dV \tag{16}$$

where the integral with V denotes the integral in each element, and $\sum$ over m denotes a summation of all elements. $[B]$, which consists of derivatives of the shape functions of each element, satisfies the following equation:

$$\{\varepsilon\} = [B] \{u_b\}$$

The piano body manufacturing process involves the following steps:

- (1) Gluing the bridge and ribs to the soundboard (manufacturing a soundboard assembly)
- (2) Gluing the soundboard assembly to the rim
- (3) Fastening a cast-iron frame to the soundboard–rim assembly with bolts (manufacturing a body)
- (4) Stringing the body.

All the above-listed steps involve geometrical nonlinearity. In addition, steps (1)–(3) involve contact nonlinearity with slip during deformation.

Because of these processes, large manufacturing stress remains in the finished piano. Notably, the natural frequency and modal damping of a structure change depending on the residual stress. For example, the axial tensile stress in a beam transversely vibrating increases the natural frequency and decreases the modal damping, whereas the axial compressive stress decreases the natural frequency and increases modal damping [15, 16]. Therefore, to more accurately simulate piano sounds, the deformation and stress of the body must be calculated considering the geometrical and contact nonlinearity involved in successive manufacturing processes. This nonlinear static deformation analysis generally requires incremental solution schemes that divide the problem into steps for calculating the displacement and updating the stiffness. Therefore, the equilibrium equation can be expressed as follows:

$$[K'_b(u_b, f_b)] \{u_b\} = \{f_b\} \tag{17}$$

Here, $[B]$ in Eq. (16) is evaluated at the current deformed geometry in each step. Moreover, additional stiffness due to initial stresses is considered in each step [21].

2.6 Body

Notably, for the geometry and stiffness matrix of the body considered here, the effect of the manufacturing processes is already incorporated. The equations of motion of the body excited with a harmonic excitation angular frequency ω are derived using FEM as follows:

$$\begin{aligned} (-\omega^2 [M_b] + [K'_b] + j[K''_b]) \{U_b(\omega)\} &= \{F_b\} \\ \text{with } \{u_b(t)\} &= \{U_b(\omega)\} e^{j\omega t}, \{f_b(t)\} = \{F_b\} e^{j\omega t} \end{aligned} \tag{18}$$

制造过程中的非线性与残余应力

式 (16) 给出基于标准 FEM 的刚度矩阵积分。钢琴琴体制造依次包括：（1）把琴桥和肋木胶合到响板，形成响板组件；（2）把响板组件胶合到外壳；（3）用螺栓把铸铁板固定到响板–外壳组件，形成琴体；（4）上弦。上述步骤都涉及几何非线性；步骤（1）–（3）还存在随变形发生滑移的接触非线性。因此成品会保留显著制造应力。残余应力会改变结构固有频率和模态阻尼：例如受拉梁的横向固有频率升高而阻尼下降，受压则相反。为了更准确模拟声音，必须在连续制造步骤中考虑几何与接触非线性。作者使用增量式非线性静力分析，逐步计算位移并更新刚度，得到依赖当前变形/载荷的平衡方程式 (17)，并在每步考虑初始应力产生的附加刚度。

2.6 琴体

后续琴体几何和刚度矩阵已经包含制造过程影响。式 (18) 是琴体在谐波激励下的 FEM 动力学方程。

where $\{u_b(t)\} \in \mathbb{C}^{N_b}$ and $\{U_b(\omega)\} \in \mathbb{C}^{N_b}$ are the displacement vector of the body and its complex amplitude vector, and $\{f_b(t)\} \in \mathbb{C}^{N_b}$ and $\{F_b\} \in \mathbb{R}^{N_b}$ are the external force vector and its complex amplitude vector, respectively. $[M_b] \in \mathbb{R}^{N_b \times N_b}$ is the mass matrix of the body, and $[K_b''] \in \mathbb{R}^{N_b \times N_b}$ is the structural damping matrix, which is expressed as follows:

$$[K_b''] = \sum_m \int_V [B]^T [D''] [B] dV$$

By eliminating the attenuation term from the equation obtained by replacing ω and $\{F_b\}$ in (18) with Ω_b and $\{0\}$, respectively, the following real eigenvalue problem can be defined:

$$(-\Omega_b^2 [M_b] + [K_b']) \{U_b(\Omega_b)\} = \{0\} \tag{19}$$

By solving this eigenvalue problem, the real eigenvalue Ω_b^2 and the corresponding real eigenvector $\{\phi_b\} \in \mathbb{R}^{N_b}$ are obtained, where Ω_b denotes the undamped natural angular frequency of the body.

When the free-interface CMS (using the component modes calculated under the condition that the body is unconstrained by the strings and air) is employed to represent the displacement of a body influenced by the motion of the strings and air, the displacement amplitude of the body $\{U_b(\omega)\}$ can be expressed as follows:

$$\{U_b(\omega)\} = [\phi_b] \{q_b(\omega)\} \tag{20}$$

where $[\phi_b] \in \mathbb{R}^{N_b \times R_b}$ and $\{q_b(\omega)\} \in \mathbb{C}^{R_b}$ are the modal matrix (the matrix where the eigenvectors obtained above are arranged horizontally) and the modal coordinate, respectively, with the number of the considered natural mode R_b ($\ll N_b$). When Eq. (20) is substituted into Eq. (18) and multiplied by $[\phi_b]^T$ from the left side of Eq. (18), the following equation is obtained:

$$[\phi_b]^T (-\omega^2 [M_b] + [K_b'] + j[K_b'']) [\phi_b] \{q(\omega)\} = [\phi_b]^T \{F_b\}$$

When $[\phi_b]$ is normalized such that $[\phi_b]^T [M_b] [\phi_b] = [I_b]$, the following equation is obtained:

$$(-\omega^2 [I_b] + [\Omega_b^2] + j[G_b]) \{q_b(\omega)\} = [\phi_b]^T \{F_b\} \tag{21}$$

$$\text{with } [G_b] = [\phi_b]^T [K_b''] [\phi_b] \tag{22}$$

where $[I_b] \in \mathbb{R}^{R_b \times R_b}$ is the identity matrix, and $[G_b] \in \mathbb{R}^{R_b \times R_b}$ is the symmetric modal structural damping matrix. $[\Omega_b^2] \in \mathbb{R}^{R_b \times R_b}$ is a diagonal matrix with $(\Omega_b^{[r_b]})^2$ ($r_b = 1, 2, \dots, R_b$) arranged diagonally, where r_b denotes the index of the normal mode of the undamped body.

2.6 琴体 (续)

$\{u_b(t)\}$ 、 $\{U_b(\omega)\}$ 分别是琴体位移向量和复振幅； $\{f_b(t)\}$ 、 $\{F_b\}$ 是外力向量及其复振幅； $[M_b]$ 是质量矩阵， $[K' _b]$ 是结构阻尼矩阵，其 FEM 积分表达式见左页。
从式 (18) 去掉衰减项，并把激励置零，可定义实特征值问题式 (19)，求得无阻尼固有角频率 Ω_b 与实特征向量 ϕ_b 。
采用自由界面 CMS（琴体不受弦和空气约束）后，位移振幅可表示为 $\{U_b\}=[\phi_b]\{q_b\}$ （式 20），其中 $R_b \ll N_b$ 。把该展开代入式 (18)，并按单位模态质量归一化，得到模态空间方程式 (21)，结构阻尼投影为对称模态阻尼矩阵 $[G_b]=[\phi_b]^T[K' _b][\phi_b]$ （式 22）。

2.7 Body–air system

Here, we consider the problem of a coupled body–air system. We deal with external problems in which a body exists in an infinite domain or a semi-infinite domain, representing an anechoic chamber or a semi-anechoic chamber, respectively. In any case, the Sommerfeld condition is imposed as the boundary condition for infinity. To solve the 3D large-scale external acoustic problem of an object with complex shapes, either the FEM with PML or boundary element method (BEM) with the fast multipole method (FMM), i.e., FMBEM [22–24] is usually employed. In our preliminary trial on frequency response analysis of an external acoustic problem with the exact solution, i.e., a problem of acoustic radiation from a circular piston on a rigid sphere [25], the FMBEM outperformed the FEM with PML in terms of both accuracy and computational cost. Therefore, FMBEM is adopted here.

When the free-interface CMS (using the component modes calculated under the condition that the body–air system is unconstrained by the strings) is employed to represent the displacement of the body–air system influenced by the motion of the strings, the boundary integral equations based on the equations of motion of the coupled body–air system excited with the harmonic excitation angular frequency ω are derived using BEM with Eq. (21) as follows [26]:

$$\begin{bmatrix} [A_a(\omega)] & \omega^2 \rho_a [B_a(\omega)][C_a][\phi_a] \\ -([C_a][\phi_a])^T [S_a] & -\omega^2 [I_b] + [\Omega_b]^2 + j[G_b] \end{bmatrix} \begin{Bmatrix} P_a(\omega) \\ q_b(\omega) \end{Bmatrix} = \begin{Bmatrix} j\omega \rho_a [D_a(\omega)] \{Q_a\} \\ [\phi_b]^T \{F_b\} \end{Bmatrix} \quad (23)$$

with $\{p_a(t)\} = \{P_a(\omega)\} e^{j\omega t}$, $\{f_b(t)\} = \{F_b\} e^{j\omega t}$

where $\{p_a(t)\} \in \mathbb{C}^{N_a}$ and $\{P_a(\omega)\} \in \mathbb{C}^{N_a}$ are the sound pressure vector on the surface of the body and its complex amplitude vector, respectively, and N_a is the number of boundary elements on the surface of the body. $[\phi_a] \in \mathbb{R}^{3N_a \times R_b}$ is the modal matrix consisting of the (X, Y, Z) component of the eigenvectors at the center of the boundary element, which is calculated by interpolating $[\phi_b]$. $[A_a(\omega)] \in \mathbb{C}^{N_a \times N_a}$ and $[B_a(\omega)] \in \mathbb{C}^{N_a \times N_a}$ are the influence coefficient matrices of the air on the body surface. $[C_a] \in \mathbb{R}^{N_a \times 3N_a}$ is the matrix comprising the direction cosine of the normal vector to the boundary element, $[S_a] \in \mathbb{R}^{N_a \times N_a}$ is a diagonal matrix comprising the boundary element area arranged diagonally, and $\{Q_a\} \in \mathbb{R}^{I_P}$ and $[D_a(\omega)] \in \mathbb{C}^{N_a \times I_P}$ are the volume velocity vector and the influence coefficient matrix at the point in the air with the number of the point source I_P . To overcome the difficulty of non-uniqueness in an external problem, $[A_a(\omega)]$, $[B_a(\omega)]$, and $[D_a(\omega)]$ are constructed using the Burton–Miller method. When the values of the right side of Eq. (23) are substituted, Eq. (23) becomes the system of linear equations with respect to the unknown variables $\{P_a(\omega)\}$ and $\{q_b(\omega)\}$. The solution to this problem, i.e., frequency response, can be obtained using an iterative method with FMM.

If the frequency response on the surface of a body surrounded by air can be expressed as a superposition of normal modes, then time-domain simulations of a coupled system consisting of nonlinear strings and a linear body–air system can be performed efficiently. The modal parameters on the surface of a body surrounded by air can be obtained by solving the following nonlinear eigenvalue problem, which is

2.7 琴体–空气系统

研究外部声学问题：琴体位于无限或半无限域，分别对应消声室或半消声室，并在无穷远施加 Sommerfeld 辐射条件。对复杂几何的大尺度三维外部声学，常用带 PML 的 FEM 或结合快速多极法 FMM 的边界元，即 FMBEM。作者以具有解析解的刚性球面圆活塞辐射问题进行预试验，发现 FMBEM 在精度和计算成本上优于 FEM+PML，因此采用 FMBEM。

对琴体–空气系统使用自由界面 CMS（系统不受琴弦约束），并以式 (21) 的琴体模态方程为基础，通过 BEM 推导谐波激励下的耦合边界积分方程式 (23)。未知量包括琴体表面声压 $P_a(\omega)$ 和琴体模态坐标 $q_b(\omega)$ 。 A_a 、 B_a 是空气–结构边界影响矩阵； C_a 包含边界法向方向余弦； S_a 是边界单元面积对角矩阵； Q_a 和 D_a 描述空气中的点体积速度源及其影响。

为解决外部 BEM 的非唯一性问题， A_a 、 B_a 、 D_a 采用 Burton–Miller 方法构造。给定右端激励后，式 (23) 是可用 FMM 迭代求解的大型线性系统。

defined by setting the right side of Eq. (23) as $\{0\}$ and replacing ω in the left side of Eq. (23) with the eigenvalue λ :

$$\begin{bmatrix} [A_a(\lambda)] & \lambda^2 \rho_a [B_a(\lambda)][C_a][\phi_a] \\ -([C_a][\phi_a])^T[S_a] & -\lambda^2[I_b] + [\Omega_b^2] + j[G_b] \end{bmatrix} \begin{Bmatrix} P_a(\lambda) \\ q_b(\lambda) \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (24)$$

This nonlinear eigenvalue problem with dense, complex, and asymmetric matrix can be solved using the contour integral method with FMM [27–31]. By applying this method, the complex eigenvalues and right complex eigenvectors with the arbitrariness of constant multiples can be obtained. To completely determine the modal parameters on the body surface surrounded by air, it is necessary to determine not only the right but also the left eigenvectors and eliminate the arbitrariness of constant multiples, i.e., normalize the eigenvectors. To solve the remaining problems and enable time-domain simulations later, following the theory of experimental mode identification, we assume that the body surrounded by air can be approximated as a general viscous damping system (non-proportional viscous damping system) with a real symmetric matrix $[M]$, $[C]$ and $[K] \in \mathbb{R}^{N_b \times N_b}$, which includes the effect of structural damping and air resistance. Because our original body model is a structural damping system that includes an imaginary number, this assumption does not hold strictly. However, when dealing with a structural damping system in the time domain, it is necessary to approximate it with a viscous damping system. Here, the equations of motion of a piano body surrounded by air are described using its displacement vector $\{u_b(t)\} \in \mathbb{R}^{N_b}$ and external force vector $\{f_b(t)\} \in \mathbb{R}^{N_b}$ as follows:

$$[M]\{\ddot{u}_b(t)\} + [C]\{\dot{u}_b(t)\} + [K]\{u_b(t)\} = \{f_b(t)\} \quad (25)$$

These equations can be transformed as follows:

$$[D]\{\dot{w}_b(t)\} + [E]\{w_b(t)\} = \{g_b(t)\} \quad (26)$$

with

$$[D] = \begin{bmatrix} C & M \\ M & 0 \end{bmatrix}, \quad [E] = \begin{bmatrix} K & 0 \\ 0 & -M \end{bmatrix}, \quad \{w_b(t)\} = \begin{Bmatrix} u_b(t) \\ \dot{u}_b(t) \end{Bmatrix}, \quad \{g_b(t)\} = \begin{Bmatrix} f_b(t) \\ 0 \end{Bmatrix}$$

Substituting $\{w_b(t)\} = \{W_b(\lambda)\}e^{\lambda t}$ and $\{g_b(t)\} = \{0\}$ into Eq (26), the following complex eigenvalue problem can be defined.

$$(\lambda[D] + [E])\{W_b(\lambda)\} = \{0\} \quad (27)$$

Considering that $[D]$ and $[E] \in \mathbb{R}^{2N_b \times 2N_b}$ are real symmetric matrices, if we solved this eigenvalue problem, the modal parameters (i.e., the complex eigenvalue λ and the complex modal matrix $[\psi] \in \mathbb{C}^{N_b \times R}$) would be obtained with the number of the considered normal mode R . Using these modal parameters, the basic solution of Eq. (25) in the frequency domain, i.e., the frequency response function (compliance)

琴体-空气复模态：非线性特征值问题

如果空气包围的琴体表面频响能够表示为正规模叠加，就可以高效地做“非线性琴弦 + 线性琴体-空气”的时域耦合。把式 (23) 右端置零，并用特征值 λ 替换频率 ω ，得到非线性特征值问题式 (24)。其矩阵是稠密、复数、非对称的，可借助 FMM 的围道积分法求解，得到复特征值与右复特征向量。

为了完整确定模态参数，还需确定左特征向量并消除特征向量任意倍数，即进行归一化。作者借鉴实验模态识别理论，近似假设空气包围的琴体可以等效为具有实对称 $[M]$ 、 $[C]$ 、 $[K]$ 的一般黏性阻尼（非比例阻尼）系统。原始琴体使用复数结构阻尼，因此该假设并不严格，但为了时域实现需要用黏性阻尼近似结构阻尼。

式 (25) 为等效二阶运动方程；式 (26) 将其改写成一阶形式；再令状态按 $e^{\lambda t}$ 变化得到线性复特征值问题式 (27)。

$H_b^{[i][l]}(\omega)$, is described as follows [32]:

$$H_b^{[i][l]}(\omega) = \frac{U_b^{[l]}(\omega)}{F_b^{[i]}} = \sum_{r=1}^R \left(\frac{Re^{[r][i][l]}}{j\omega - \lambda^{[r]}} + \frac{\bar{R}e^{[r][i][l]}}{j\omega - \bar{\lambda}^{[r]}} \right) \quad (28)$$

with $u_b^{[l]}(t) = U_b^{[l]}(\omega) e^{j\omega t}$, $f_b^{[i]}(t) = F_b^{[i]} e^{j\omega t}$, $\lambda^{[r]} = -\sigma^{[r]} + j\omega_d^{[r]}$, $Re^{[r][i][l]} = \psi^{[r][i]} \psi^{[r][l]}$

where $Re^{[r][i][l]}$ is the modal residue, $\sigma^{[r]}$ and $\omega_d^{[r]}$ are the modal decay rate and damped natural angular frequency, $\psi^{[r][i]}$ and $\psi^{[r][l]}$ are the components of the normalized complex eigenvector at points i and l , and r, i and l are the indices of the normal mode, excitation and response points, respectively. The bar above a character indicates the conjugate of a complex number.

Notably, the aforementioned assumption holds that the modal parameters on the surface of the body obtained by Eq. (27) should be approximately equal to those obtained using Eq. (24). In addition, Eq. (28) derived from Eq. (27) indicates that the left and right eigenvectors are the same. Thus, normalized complex eigenvectors can be determined by solving the following linear least squares problem:

$$\begin{aligned} & \text{Minimize} \quad \sum_{k,i,l} \{Er^{[k][i][l]}(z^{[r]})\}^2 \\ & Er^{[k][i][l]}(z^{[r]}) = \sum_{r=1}^R \left(\frac{Re^{[r][i][l]}}{j\omega^{[k]} - \lambda^{[r]}} + \frac{\bar{R}e^{[r][i][l]}}{j\omega^{[k]} - \bar{\lambda}^{[r]}} \right) - H_b^{[i][l]}(\omega^{[k]}) \end{aligned} \quad (29)$$

with $\lambda^{[r]} = -\sigma^{[r]} + j\omega_d^{[r]}$, $Re^{[r][i][l]} = \psi^{[r][i]} \psi^{[r][l]}$, $\psi^{[r][i]} = z^{[r]} \psi_T^{[r][i]}$, $\psi^{[r][l]} = z^{[r]} \psi_T^{[r][l]}$

where $\lambda^{[r]}$, $\psi_T^{[r][i]}$, and $\psi_T^{[r][l]}$ are the eigenvalue and components of the non-normalized eigenvector at points i and l , respectively. They are obtained by the nonlinear eigenvalue analysis of Eq. (24), considering Eq. (20). $H_b^{[i][l]}(\omega^{[k]})$ is the frequency response (compliance) obtained by the frequency response analysis of Eq. (23) with $\{Q_a\} = \{0\}$ and $F_b^{[i]} = 1$ at some representative points extracted from all the string support points, considering Eq. (20). k is the index for discrete frequency, and $z^{[r]}$ is the complex decision variable. Notably, the matrices $[M]$, $[C]$, $[K]$, $[D]$ and $[E]$ in Eqs. (25)–(27) are only used for theoretical considerations and are not calculated in the proposed method.

When the complex CMS is applied to the coupled body–air system using the determined modal parameters, Eq. (28) yields the displacement expression using the complex modal coordinate $\eta^{[r]}(t)$ ($r = 1, 2, \dots, R$) and linear ordinary differential equations with respect to the complex modal coordinate as follows:

$$u_b^{[l]}(t) = 2 \sum_{r=1}^R \Re \left[\psi^{[r][l]} \eta^{[r]}(t) \right] \quad (30)$$

$$\dot{\eta}^{[r]}(t) - \lambda^{[r]} \eta^{[r]}(t) = \sum_i \psi^{[r][i]} f_b^{[i]}(t) \quad r = 1, 2, \dots, R \quad (31)$$

复模态频响与归一化

若求得式 (27) 的复特征值 $\lambda^{[r]}$ 和复模态矩阵 ψ , 则琴体的频率响应 (柔度) 可表示为式 (28): 每个复模态贡献一对共轭极点项。 $\lambda^{[r]} = -\sigma^{[r]} + j\omega_d^{[r]}$, 其中 $\sigma^{[r]}$ 为模态衰减率, $\omega_d^{[r]}$ 为阻尼固有角频率; 模态残数 $Re^{[r][i][l]} = \psi^{[r][i]} \psi^{[r][l]}$ 。 i, l 分别是激励点和响应点索引。

理论假设要求式 (27) 获得的表面模态参数应近似等于式 (24) 直接获得的参数; 式 (28) 还表明左右特征向量相同。因此作者通过线性最小二乘问题式 (29) 求归一化系数, 使由复极点/残数重建的频响在代表性弦支承点尽量拟合式 (23) 的直接 FMBEM 频率响应。

式 (25)–(27) 中的等效 $[M]$ 、 $[C]$ 、 $[K]$ 只用于理论推导, 本文方法并不显式计算这些矩阵。

Furthermore, when the displacement $u_b^{[l]}$ and external force $f_b^{[i]}(t)$ are replaced with the displacement of the string end-support $u_{Bs}^{[i_B]}(t)$ ($s = z, x, y$) in Eq. (7) and the force exerted by the strings $f_{Bs}^{[i_B]}(t)$ ($s = z, x, y$) in Eq. (13) respectively, Eqs. (30) and (31) yield the following equations:

$$u_{Bs}^{[i_K][i_W][i_L][i_B]}(t) = 2 \sum_{r=1}^R \Re \left[\hat{\psi}_{Bs}^{[r][i_K][i_W][i_L][i_B]} \eta^{[r]}(t) \right] \\ i_K = 1, 2, \dots, I_K; \ i_W = 1, 2, \dots, I_W^{[i_K]}; \ i_L = 1, 2; \ i_B = 0, 1; \ s = z, x, y \quad (32)$$

$$\dot{\eta}^{[r]}(t) - \lambda^{[r]} \eta^{[r]}(t) = \sum_{i_K=1}^{I_K} \sum_{i_W=1}^{I_W^{[i_K]}} \sum_{i_L=1}^2 \sum_{i_B=0}^1 \sum_{s=z,x,y} \hat{\psi}_{Bs}^{[r][i_K][i_W][i_L][i_B]} f_{Bs}^{[i_K][i_W][i_L][i_B]}(t) \\ \text{with} \quad \hat{\psi}_{Bs}^{[r][i_K][i_W][i_L][i_B]} = \sum_{S=Z,X,Y} \beta_{Ss}^{[i_K][i_W][i_L][i_B]} \psi_{Bs}^{[r][i_K][i_W][i_L][i_B]} \quad r = 1, 2, \dots, R \quad (33)$$

where i_K and I_K ($= 88$ for general grand pianos) are the index and number of the key, respectively. $\beta_{Ss}^{[i_K][i_W][i_L][i_B]}$ are the direction cosine between the XYZ - and xyz -coordinates, and $\psi_{Bs}^{[r][i_K][i_W][i_L][i_B]}$ are the components of the normal mode at the string support point on the XYZ -coordinate. The moment of force $m_{Bs}^{[i_K][i_W][i_L][i_B]}(t)$ in Eq. (13) can be approximately considered as a couple component of the force acting on two adjacent string support points. In the same way, the rotation angle $\theta_{Bs}^{[i_K][i_W][i_L][i_B]}(t)$ in Eq. (7) can be approximately considered as a rotation of the line defined by the two adjacent string support points.

2.8 Whole system

The equation of motion of the hammer Eq. (1), the equation of equilibrium of forces on the hammer felt Eq. (2), the equation of motion of each mode of the string (12), and the equation of motion of each mode of the body surrounded by air (33) are collectively referred to as the equation of motion of the whole system. Considering that the variables that represent the interactions between the substructures (i.e., $f_{Ha}^{[i_K][i_W]}(t)$, $f_{Hb}^{[i_K]}(t)$, $u_s^{[i_K][i_W]}(x_H^{[i_K][i_W]}, t)$, $f_{Bs}^{[i_K][i_W][i_L][i_B]}(t)$, $m_{Bs}^{[i_K][i_W][i_L][i_B]}(t)$, $u_{Bs}^{[i_K][i_W][i_L][i_B]}(t)$, and $\theta_{Bs}^{[i_K][i_W][i_L][i_B]}(t)$) are erasable intermediate variables, the equation of motion of the whole system can be expressed as simultaneous nonlinear ordinary differential equations with respect to the unknown variables $u_H^{[i_K]}(t)$, $u_{Hab}^{[i_K]}(t)$, $\xi_{zy}^{[r_{zy}][i_K][i_W][i_L]}(t)$, and $\xi_x^{[r_x][i_K][i_W][i_L]}(t)$ ($i_K = 1, 2, \dots, I_K$; $i_W = 1, 2, \dots, I_W^{[i_K]}$; $i_L = 1, 2$; $r_{zy} = 1, 2, \dots, R_{zy}^{[i_K][i_L]}$; $r_x = 1, 2, \dots, R_x^{[i_K][i_L]}$); and $\eta^{[r]}(t)$ ($r = 1, 2, \dots, R$).

By setting the status of the piano at rest as the initial condition, the problem can be considered as the initial value problem of the simultaneous nonlinear ordinary differential equations. The motion of the whole system is created when the input to the system (the hammer's initial velocity) is given.

确定复模态参数后, 式 (30) 用复模态坐标 $\eta^{[r]}(t)$ 表示琴体位移, 式 (31) 给出每个复模态的一阶线性 ODE。

把琴体响应点替换为各根弦的琴桥/弦枕端, 把外力替换为弦在支承端施加的三维力, 即得到式 (32)、(33): 每个支承点的三维位移由所有琴体-空气复模态叠加得到; 每个复模态由全部 88 键、每键各根弦、有效弦段/后弦段、两端支承及三个空间方向的弦端力共同激励。局部弦坐标与全局琴体坐标之间用方向余弦转换。弦端弯矩可近似为作用于相邻两个支承点的一对力偶; 支承端转角可近似为相邻支承点连线的转动。

2.8 完整系统

击锤式 (1)、毡力平衡式 (2)、弦各模态运动式 (12)、空气包围琴体的复模态运动式 (33) 共同构成整机运动方程。消去各子系统之间的中间交互变量后, 得到以击锤位移、毡层连接位移、各弦横向/纵向模态坐标和琴体-空气复模态坐标为未知量的联立非线性 ODE 初值问题。以钢琴静止为初态, 只要给定输入——击锤初速度——整机运动就被激发。

2.9 Sound radiation

The sound pressure at any observation point in the air $p^{[ip]}(t)$ is calculated by a convolution operation as follows:

$$p^{[ip]}(t) = \sum_{i_K=1}^{I_K} \sum_{i_W=1}^{I_W^{[i_K]}} \sum_{i_L=1}^2 \sum_{i_B=0}^1 \sum_{s=z,x,y} \int_0^t f_{Bs}^{[i_K][i_W][i_L][i_B]}(t-\tau) h_s^{[i_K][i_W][i_L][i_B][i_P]}(\tau) d\tau$$
$$i_P = 1, 2, \cdots, I_P \tag{34}$$
$$\text{with } h_s^{[i_K][i_W][i_L][i_B][i_P]}(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} H_s^{[i_K][i_W][i_L][i_B][i_P]}(\omega) e^{j\omega t} d\omega$$

where $H_s^{[i_K][i_W][i_L][i_B][i_P]}(\omega)$ is the frequency response function between the force exerted by the string at the string support point and the sound pressure at the observation point in air (the virtual microphone point), and i_P and I_P are the index and number of the virtual microphone points, respectively. When calculating $H_s^{[i_K][i_W][i_L][i_B][i_P]}(\omega)$, considering that the string support points (with 3D direction) are usually much more than the virtual microphone points, it is effective to use the reciprocity principle for structural-acoustic coupling problems as follows:

$$H_s^{[i_K][i_W][i_L][i_B][i_P]}(\omega) = \frac{P^{[i_P]}(\omega)}{F_{Bs}^{[i_K][i_W][i_L][i_B]}} = \frac{j\omega U_{Bs}^{[i_K][i_W][i_L][i_B]}(\omega)}{Q_a^{[i_P]}} \tag{35}$$
$$\text{with } u_{Bs}^{[i_K][i_W][i_L][i_B]}(t) = U_{Bs}^{[i_K][i_W][i_L][i_B]}(\omega) e^{j\omega t}, \quad f_{Bs}^{[i_K][i_W][i_L][i_B]}(t) = F_{Bs}^{[i_K][i_W][i_L][i_B]} e^{j\omega t}$$

This principle states that the sound pressure at point A in the air exerted by the force vector at point B on the structural surface is equal to the velocity vector at point B on the structural surface exerted by the volume velocity at point A in the air. $H_s^{[i_K][i_W][i_L][i_B][i_P]}(\omega)$ are obtained by solving the system of equations (23) with $Q_a^{[i_P]} = 1$ and $\{F_b\} = \{0\}$ at each frequency, considering Eq. (20).

3 Numerical analysis

A 3D model of the Yamaha concert grand piano CFX (W = 1.60 m, H = 1.85 m, D = 2.75 m) with an open top lid was constructed (Fig. 8). Considering the calculation accuracy, a hexahedral mesh rather than a tetrahedral mesh was applied to most parts. On this piano, I_K is 88, I_W is 1 from the 1st key (note A1) to the 8th key (note E1), 2 from the 9th key (note F1) to the 13th key (note A2), and 3 from the 14th key (note A#2) to the 88th key (note C8). Wrapped strings are incorporated from the 1st key (note A1) to the 20th key (note E2), whereas plain strings are incorporated from the 21st key (note F2) to the 88th key (note C8).

3.1 Preprocesses

Processes preceding the time-domain simulation are referred to as preprocesses.

2.9 声辐射

任一空气观察点（虚拟麦克风）的声压 $\hat{p}[\text{iP}](t)$ 由式（34）的卷积求得：把所有琴键、所有同音弦、有效/后弦段、两端支承以及三个空间方向上的弦端力，与对应“弦支承点力 $\rightarrow$ 虚拟麦克风声压”的脉冲响应 h_s 卷积，再全部求和。

$H_s(\omega)$ 是弦端力到观察点声压的频率响应。由于三维弦支承点数量远大于虚拟麦克风数量，作者利用结构-声耦合的互易原理（式 35）：空气中 A 点由结构表面 B 点力产生的声压，等价于在 A 点施加体积速度源时结构表面 B 点产生的速度响应。这样只需对每个虚拟麦克风源进行 FMBEM 频响分析，就能得到所有弦端到该麦克风的传递函数。

3 数值分析

建立 Yamaha CFX 音乐会三角钢琴三维模型（宽 1.60 m、高 1.85 m、深 2.75 m，顶盖开启）。多数部件使用六面体而非四面体网格以提高精度。88 键中 A1–E1 每键 1 根弦，F1–A2 每键 2 根弦，A#2–C8 每键 3 根弦；A1–E2 使用缠弦，F2–C8 使用裸弦。

Table 1 Physical properties of the hammer. ($I_{Hb} = 2$)

Physical properties		A1	A5	C8	Units
Hammer mass	M_H	13.2	9.12	5.33	g
Stiffness coefficient (upper)	K_{Ha}	5.37×10^2	2.83×10^3	5.10×10^4	$\text{N}/\text{mm}^{p_{Ha}}$
Nonlinear exponent (upper)	p_{Ha}	4.20	5.69	4.50	
Relaxation time (upper)	τ_{Ha}	1.0×10^{-5}	1.0×10^{-5}	1.0×10^{-5}	s
Positive real number (upper)	β_{Ha}	0.80	0.80	0.80	
Stiffness coefficient (lower)	K_{Hb}	8.02×10^3	8.52×10^3	7.86×10^4	$\text{N}/\text{mm}^{p_{Hb}}$
Nonlinear exponent (lower)	p_{Hb}	2.00	2.00	1.50	
Relaxation time (lower)	$\tau_{Hb}^{[1]}$	1.0×10^{-5}	1.0×10^{-5}	1.0×10^{-5}	s
Positive real number (lower)	$\beta_{Hb}^{[1]}$	0.50	0.50	0.50	
Relaxation time (lower)	$\tau_{Hb}^{[2]}$	5.0×10^{-6}	5.0×10^{-6}	5.0×10^{-6}	s
Positive real number (lower)	$\beta_{Hb}^{[2]}$	0.20	0.20	0.20	
Relative striking position	$1/2 - x_H/L$	0.126	0.123	0.054	

The preprocesses for hammers and strings include identifying the nonlinear vis-coelasticity properties of the hammer felt and eigenvalue analysis (calculation of the eigenvalues and eigenmodes) of the strings.

The properties of the double-layer nonlinear generalized Maxwell model for the hammer felt were identified using the differential evolution algorithm [33] so that the spectrum of the final synthesized sound would be as close as possible to a piano recording. This approach is analogous to how piano technicians bring the output closer to an ideal sound in their imagination. Table 1 lists the values of the physical properties used for the hammers.

The initial normal strain of the strings ε_0 and the natural angular frequencies of the transverse vibration of the strings $\Omega_{zy}^{[r_{zy}]}$ ($r_{zy} = 2, 3, \dots, R_{zy}$) in Eqs. (9a) and (9b) are determined by numerically solving the system of transcendental equations with quadruple precision. $A_s^{[r_{zy}]}, B_s^{[r_{zy}]}, C_s^{[r_{zy}]}, D_s^{[r_{zy}]}$ ($r_{zy} = 1, 2, \dots, R_{zy}; s = z, y$) in Eqs. (9a) and (9b) are determined by solving the simultaneous linear equations with $A_z^{[r_{zy}]} = 1$ and then normalizing the modal mass to unity. Tables 2 and 3 list the physical properties used for the strings and ends of the strings, respectively, and Table 5 lists the numerical parameters used for the strings. The attenuation properties η'_{zy} , η'_x , β' , p' and the string-ends properties $K_{Bz'}^{[0]}, K_{By'}^{[0]}, K_{Bz'}^{[1]}, K_{By'}^{[1]}$ were chosen so that the synthesized sound would be as realistic as possible, considering that the rotation of the string at its support ends is somewhere between “constrained” and “unconstrained.” Due to the anisotropy of the string end supports, pairs of close angular frequencies occur in the natural angular frequency sequence of bending vibration. In the future, by focusing on these frequency differences, it is expected that the string-ends properties $K_{Bz'}^{[0]}, K_{By'}^{[0]}, K_{Bz'}^{[1]}$, and $K_{By'}^{[1]}$ can be adjusted more precisely. The normal modes of the strings were considered up to approximately 10 kHz (for the low range), 12 kHz (for the mid-range), and 14 kHz (for the high range).

Table 4 lists the values of the physical properties used for the dampers.

3.1 预处理：击锤与弦参数

表 1：击锤物理参数（下层 Maxwe11 支路数 I_Hb=2）。左页列出 A1、A5、C8 三个代表音的锤重、上下层刚度系数、非线性指数、松弛时间、 β 权重，以及相对击弦位置。

击锤毡双层非线性广义 Maxwell 模型的参数通过差分进化算法识别，使最终合成声音的频谱尽量接近钢琴实录。作者把它类比为钢琴技师通过整音把输出逼近心中理想声音。

弦的初始正应变 ε_0 与横向固有角频率通过四倍精度数值求解超越方程组确定；式（9）模态系数通过线性方程求得并将模态质量归一化为 1。表 2、3 给出弦和弦端物理参数，表 5 给出弦的数值参数。

衰减参数 η 、 β 、p，以及两端转动弹簧刚度，是以“使合成声音尽可能真实”为目标选择的。端部各向异性会使弯曲固有频率序列出现成对的近邻频率；未来可利用这些频差更精确标定端部参数。弦模态截断到约：低音 10 kHz、中音 12 kHz、高音 14 kHz。

Table 2 Physical properties of the string.

Physical properties		A1	A5	C8	Units
Density of core wire	ρ_f	7.86×10^3	7.86×10^3	7.86×10^3	kg/m ³
Density of copper wire	ρ_c	8.82×10^3			kg/m ³
Modulus of longitudinal elasticity	E	200	200	200	GPa
Modulus of transverse elasticity	G	80	80	80	GPa
Timoshenko shear coefficient	k	0.9	0.9	0.9	
Transverse structural damping coefficient	η'_{zy}	1.0×10^{-5}	1.0×10^{-5}	1.0×10^{-5}	
Longitudinal structural damping coefficient	η'_x	1.0×10^{-3}	5.0×10^{-1}	5.0×10^{-1}	
Damping correction coefficient	β'	2.0×10^{-10}	2.0×10^{-11}	3.0×10^{-11}	s ^{p'}
Damping correction exponent	p'	1.50	1.54	1.60	
String length (speaking part)	$L^{[1]}$	2.018	0.405	0.052	m
String length (rear duplex part)	$L^{[2]}$	0.162	0.097	0.037	m
Diameter of core wire	d_f	1.5×10^{-3}	9.8×10^{-4}	8.0×10^{-4}	m
Diameter of copper wire	d_c	1.7×10^{-3}			m
Fundamental frequency	$\Omega^{[1]}_{zy}/(2\pi)$	27.53	442.0	4323	Hz

Table 3 Physical properties of the string end support.

Physical properties		A1	A5	C8	Units
Bridge pin angle (bridge)	$\alpha^{[0]}$	21.5	21.5	21.5	deg
Rotational spring constant (bridge)	$K^{[0]}_{Bz'}$	40	40	20	N·m
Rotational spring constant (bridge)	$K^{[0]}_{By'}$	80	80	40	N·m
bridge pin angle (iron frame)	$\alpha^{[1]}$	0.0	0.0	0.0	deg
rotational spring constant (iron frame)	$K^{[1]}_{Bz'}$	60	60	30	N·m
Rotational spring constant (iron frame)	$K^{[1]}_{By'}$	30	30	10	N·m

Table 4 Physical properties of the damper

Physical properties		A1	A5	C8	Units
Number of damper tips on each string	I_D	6	6	0	
Loss elastic coefficient	K''_D	2.42×10^3	2.42×10^3		kg/s ²
Nonlinear exponent	p_D	2.0	2.0		
Damper length	L_D	8.85×10^{-2}	5.52×10^{-2}		m

Table 5 Numerical parameters for the string. ($I_K = 88$)

Numerical parameters		A1	A5	C8
Index of key	i_K	1	49	88
Number of strings in each key	I_W	1	3	3
Number of natural modes (speaking part)	$R^{[1]}_{zy}$	410	46	6
Number of natural modes (speaking part)	$R^{[1]}_x$	24	3	0
Number of natural modes (rear duplex part)	$R^{[2]}_{zy}$	16	10	4
Number of natural modes (rear duplex part)	$R^{[2]}_x$	0	0	0

表 2-5: 弦、弦端、制音器与模态数量

表 2 (弦物理参数): 芯弦钢密度 7.86×10^3 kg/m ; 铜绕线密度 8.82×10^3 kg/m ; 纵向弹性模量 $E=200$ GPa; 剪切/横向弹性模量 $G=80$ GPa; Timoshenko 剪切系数 $k=0.9$ 。还列出横向/纵向结构阻尼、阻尼修正参数、有效弦长与后弦段长度、芯弦/铜线直径和基频。代表值: A1 有效弦长 2.018 m, A5 0.405 m, C8 0.052 m。

表 3 (弦端支承): 琴桥桥钉倾角统一为 21.5° ; 列出琴桥与铸铁板两端、两个主方向的转动弹簧常数。

表 4 (制音器): A1/A5 每根弦 6 个制音接触点, C8 无制音器; 列出损耗弹性系数、非线性指数和制音器长度。

表 5 (模态数量): A1 有效弦段保留 410 个横向模态和 24 个纵向模态, 后弦段 16 个横向模态; A5 为 46/3/10; C8 为 6/0/4。由此可见低音弦模型的状态维数显著高于高音。

Table 6 Physical properties of wood materials used in the piano body.

Physical properties		Spruce	Maple	Beech	Units
Density	ρ	0.45×10^3	0.69×10^3	0.62×10^3	kg/m^3
Modulus of longitudinal elasticity	E'_L	13.8	15.6	12.3	GPa
Modulus of longitudinal elasticity	E'_R	0.94	2.03	1.32	GPa
Modulus of longitudinal elasticity	E'_T	0.51	0.77	0.59	GPa
Modulus of transverse elasticity	G'_{LR}	0.85	1.71	0.98	GPa
Modulus of transverse elasticity	G'_{RT}	0.07	0.50	0.20	GPa
Modulus of transverse elasticity	G'_{TL}	0.78	1.25	0.64	GPa
Poisson's ratio	ν_{LR}	0.37	0.42	0.40	
Poisson's ratio	ν_{RT}	0.44	0.74	0.65	
Poisson's ratio	ν_{TL}	1.7×10^{-2}	3.4×10^{-2}	2.4×10^{-2}	
Loss factor	η_L	0.64×10^{-2}	0.74×10^{-2}	1.00×10^{-2}	
Loss factor	η_R	1.76×10^{-2}	2.08×10^{-2}	2.06×10^{-2}	
Loss factor	η_T	1.76×10^{-2}	2.08×10^{-2}	2.06×10^{-2}	
Loss factor	η_{LR}	1.62×10^{-2}	1.63×10^{-2}	1.96×10^{-2}	
Loss factor	η_{RT}	4.14×10^{-2}	4.14×10^{-2}	3.90×10^{-2}	
Loss factor	η_{TL}	1.79×10^{-2}	1.87×10^{-2}	2.28×10^{-2}	

The preprocesses for the coupled body–air system reduce the physical information about the large-scale linear coupled system into complex modal parameters at the string support points of the body–air system and the impulse response between the force at the string support points and the sound pressure at the virtual microphone points. The preprocesses for the coupled body–air system are described as follows:

- 1) 3D modeling and meshing of the body
- 2) Nonlinear static analysis of the body using FEM (manufacturing process simulation)
- 3) Real eigenvalue analysis of the body using FEM
- 4) Calculation of the modal damping matrix of the body
- 5) Nonlinear eigenvalue analysis of the coupled body–air system using FMBEM
- 6) Frequency response analysis of the coupled body–air system using FMBEM (calculation of frequency response between the force and displacement at the string support points)
- 7) Normal mode identification of the coupled body–air system (normalizing the eigenvectors at the string support points)
- 8) Frequency response analysis of the coupled body–air system with FMBEM (calculation of frequency response between the force at the string support points and the sound pressure at the virtual microphone points)

Preprocess 1 involves defining materials that constitute the body. Tables 6 and 7 list the physical properties of the woods and metals used for the piano body, respectively. Spruce is used for the soundboard, ribs, braces, and key bed, whereas maple and beech are used for the bridge, rim, pin block, and legs. Brass and stainless steel are used for the casters and bridge pins, respectively. For wood properties, the average values obtained from vibration tests on a number of specimens were used [34].

表 6 与琴体–空气预处理流程

表 6 给出琴体木材的物理参数：云杉、枫木、山毛榉的密度、L/R/T 三主方向纵向弹性模量、三个剪切模量、Poisson 比以及各方向损耗因子。云杉用于响板、肋木、撑木和键床；枫木与山毛榉用于琴桥、外壳、弦轴板、琴腿等。木材参数采用多件试样振动试验的平均值。

大型线性琴体–空气系统的预处理将其全部物理信息压缩为：弦支承点处的复模态参数，以及“弦端力 → 虚拟麦克风声压”的脉冲响应。具体流程：1) 三维琴体建模和网格；2) FEM 非线性静力分析模拟制造；3) FEM 实特征值分析；4) 琴体模态阻尼矩阵；5) FMBEM 琴体–空气非线性特征值分析；6) FMBEM 弦支承点力–位移频响；7) 耦合系统正规模识别与支承点特征向量归一化；8) FMBEM 计算弦端力到虚拟麦克风声压的频响。

Table 7 Physical properties of metal materials used in the piano body.

Physical properties		Cast-iron	Brass	Stainless steel	Units
Density	ρ	7.20×10^3	8.40×10^3	7.93×10^3	kg/m^3
Modulus of longitudinal elasticity	E'	131	103	197	GPa
Modulus of transverse elasticity	G'	50.4	39.6	74.0	GPa
Poisson's ratio	ν	0.30	0.30	0.33	
Loss factor	η	2.0×10^{-3}	1.0×10^{-4}	2.0×10^{-4}	

Table 8 Numerical parameters of the finite-element body model.

Numerical parameters		
Degree of freedom	N_b	3.5×10^7
Number of normal modes	R_b	1.9×10^4

Table 9 Physical properties of air. (assumed air temperature = 15 °C)

Physical properties			Units
Density	ρ_a	1.225	kg/m^3
Speed of sound	c_a	340	m/s
Kinematic viscosity coefficient	η_a	1.5×10^{-5}	m^2/s

Preprocesses 2 and 3 (the nonlinear static and real eigenvalue analysis) described by Eqs. (17) and (19) are performed using MSC Nastran SOL400 and MSC Nastran SOL103, respectively. Preprocess 4, described by Eq. (22), is calculated using MSC Nastran DMAP (Direct Matrix Abstraction Program). Table 8 lists the numerical parameters used for the body. The normal modes of the body are considered up to 11 kHz.

For Preprocesses 5 (the nonlinear eigenvalue analysis described by Eq. (24)), 6, and 8 (frequency response analysis described by Eq. (23)), Yamaha’s in-house program named SoNATA (Software of Numerical AcousTic Analysis) is employed. In these analyses, the entire frequency range was divided into two: low-frequency (0–5 kHz) and high-frequency (5–12 kHz) ranges. To reduce computational costs, we ignored the force exerted on the body surface by air in the high-frequency range. This means replacing the lower-left submatrix of the left-side matrix in Eqs. (23) and (24) with 0. With future enhancements to our computational resources, we will no longer require this compromise for high frequency ranges. Tables 9 and 10 list the physical properties and numerical parameters of the coupled body–air system, respectively.

表 7–9 与大尺度模型参数

表 7 给出铸铁、黄铜、不锈钢的密度、弹性模量、剪切模量、Poisson 比和损耗因子。黄铜用于脚轮，不锈钢用于桥钉等。

制造非线性静力分析和琴体实特征值分析分别使用 MSC Nastran SOL400 与 SOL103；模态阻尼矩阵使用 Nastran DMAP 计算。表 8：琴体 FEM 自由度 $N_b=3.5\times10^7$ （约 3500 万），保留实模态 $R_b=1.9\times10^4$ （约 1.9 万），模态覆盖至 11 kHz。

FMBEM 的非线性特征值分析与频响分析使用 Yamaha 自研 SoNATA。全频带分为 0–5 kHz 与 5–12 kHz 两段。为降低成本，高频段忽略空气施加到琴体表面的反馈力，即把式 (23)、(24) 左下耦合子矩阵置零；作者计划未来在算力允许时取消这一近似。

表 9：空气按 15° C， $\rho_a=1.225\text{ kg/m}^3$ ，声速 340 m/s，运动黏度 $1.5\times10^{-5}\text{ m}^2/\text{s}$ 。

Table 10 Numerical parameters of the coupled body–air system.

Numerical parameters		Low-frequency range	High-frequency range
Degree of freedom	N_a	5.0×10^5	2.0×10^6
Number of normal modes	R_a	1.4×10^4	1.3×10^4
Number of virtual microphone points	I_P	2	2

3.2 Time-domain simulation

To solve the initial value problem of the simultaneous nonlinear ordinary differential equations derived in Section 2, the trapezoidal rule, which is an implicit differential scheme with second-order accuracy, is employed. To represent the unknown variables in simultaneous nonlinear ordinary differential equations as $\{u(t)\}$, this scheme converts $\{\ddot{u}(t)\}, \{\dot{u}(t)\}, \{u(t)\}$ on the continuous time to the numerical expression by $\{u[k+1]\}, \{u[k]\}, \{u[k-1]\}$ on the discrete-time as follows:

$$\begin{aligned}\{\ddot{u}(t)\} &\rightarrow (\{u[k+1]\} - 2\{u[k]\} + \{u[k-1]\})/(\Delta t)^2 \\ \{\dot{u}(t)\} &\rightarrow (\{u[k+1]\} - \{u[k-1]\})/(2\Delta t) \\ \{u(t)\} &\rightarrow (\{u[k+1]\} + 2\{u[k]\} + \{u[k-1]\})/4 \quad \text{with } t = k\Delta t; k = 0, 1, \dots\end{aligned}$$

where k is the index of time, and Δt is the time step. Considering that the frequency accuracy is important in musical tones, this scheme is applied with prewarping to the natural angular frequency of the string and the body–air system as follows:

$$\Omega_{pw} = (2/\Delta t) \tan(\Omega_{org} \Delta t/2)$$

where Ω_{org} denotes $\Omega_{zy}^{[r_{zy}]} (r_{zy} = 1, 2, \dots, R_{zy}), \Omega_x^{[r_x]} (r_x = 1, 2, \dots, R_x)$ or $\Omega^{[r]} (= \{(\sigma^{[r]})^2 + (\omega_d^{[r]})^2\}^{\frac{1}{2}}; r = 1, 2, \dots, R)$. Ω_{pw} denotes the prewarped natural angular frequency used in the time-domain implementation.

The integrals in Eqs. (4) and (5) are calculated using the Gauss–Legendre quadrature rule. The initial value problem of the simultaneous nonlinear ordinary differential equations is converted to a problem to sequentially solve the simultaneous nonlinear algebraic equations with respect to $\{u[k+1]\} (k = 0, 1, \dots)$ on the discrete-time. These simultaneous nonlinear algebraic equations are solved using the block Newton and Gauss–Seidel methods with $\Delta t = 1/88,200s (\simeq 11.34\mu s)$.

The convolution of Eq. (34) is conducted in the frequency domain using a fast convolution algorithm.

The developed programs for synthesizing the sound of a piano were named PARTITA (Programs for ART of InTerAction). This is because the essence of music art produced by a musical instrument lies in the interactions among every constructive part, and the programs calculate the interaction of the entire system.

表 10; 3.2 时域仿真

表 10: 琴体–空气系统低频段空气边界自由度 $N_a=5.0\times10^5$ 、复模态约 1.4×10^4 ; 高频段 $N_a=2.0\times10^6$ 、复模态约 1.3×10^4 ; 虚拟麦克风数量均为 2。

为求解第 2 节得到的联立非线性 ODE 初值问题，作者采用二阶精度的隐式梯形积分格式。连续时间的 $\ddot{u}$ 、 $\dot{u}$ 、 u 分别由相邻三个离散时刻 $u[k-1]$ ， $u[k]$ ， $u[k+1]$ 的中心差分/平均式替代，具体公式见左页。

由于音乐音高对频率误差非常敏感，所有弦和琴体–空气模态的固有角频率都进行 prewarping: $\Omega_{pw}=(2/\Delta t)\tan(\Omega_{org} \Delta t/2)$ 。式 (4)、(5) 的积分用 Gauss–Legendre 求积。

每个时间步需要求解关于 $u[k+1]$ 的联立非线性代数方程，使用 block Newton + Gauss–Seidel。时间步 $\Delta t=1/88,200 \text{ s}\simeq11.34 \text{ }\mu\text{s}$ 。声辐射卷积在频域用快速卷积完成。整套程序命名为 PARTITA (Programs for ART of InTerAction)，强调声音艺术来自各部件之间的交互。

4 Results

4.1 Preprocesses

Fig. 9 shows the contour plots of the signed von Mises stress obtained by the non-linear static analysis considering large deformations and contacts that simulate the manufacturing process. The total tension of all strings during the stringing (approximately 190 kN) generated large stress especially in the cast-iron frame. At each process, it was checked that the shape of the soundboard crown reproduced that of an actual piano with sufficient accuracy. Fig. 10 shows the real normal modes obtained from the real eigenvalue analysis of the body. Approximately 19,000 real eigenvalues were extracted in the frequency range of 0–11 kHz. Fig. 11 shows the complex eigenmodes obtained from the nonlinear eigenvalue analysis of the coupled body–air system. Approximately 14,000 complex eigenvalues were extracted in the region on the complex plane in a frequency range of 0–5 kHz and the range of the modal decay rate divided by 2π between 0 s^{-1} and 120 s^{-1} .

4 结果; 4.1 预处理

图 9 给出模拟制造大变形与接触过程后的有符号 von Mises 残余应力分布。上弦时全部琴弦总张力约 190 kN, 在铸铁板尤其产生巨大应力。每一个制造阶段都检查响板 crown 的形状, 并确认与真实钢琴足够一致。图 10 给出琴体实特征值分析得到的代表性实正规模。在 0–11 kHz 内提取约 19,000 个实特征值。图 11 给出琴体-空气耦合系统非线性特征值分析得到的复特征模态。在 0–5 kHz 且模态衰减率 $\sigma/(2\pi)$ 位于 0–120 s⁻¹ 的复平面区域内, 提取约 14,000 个复特征值。

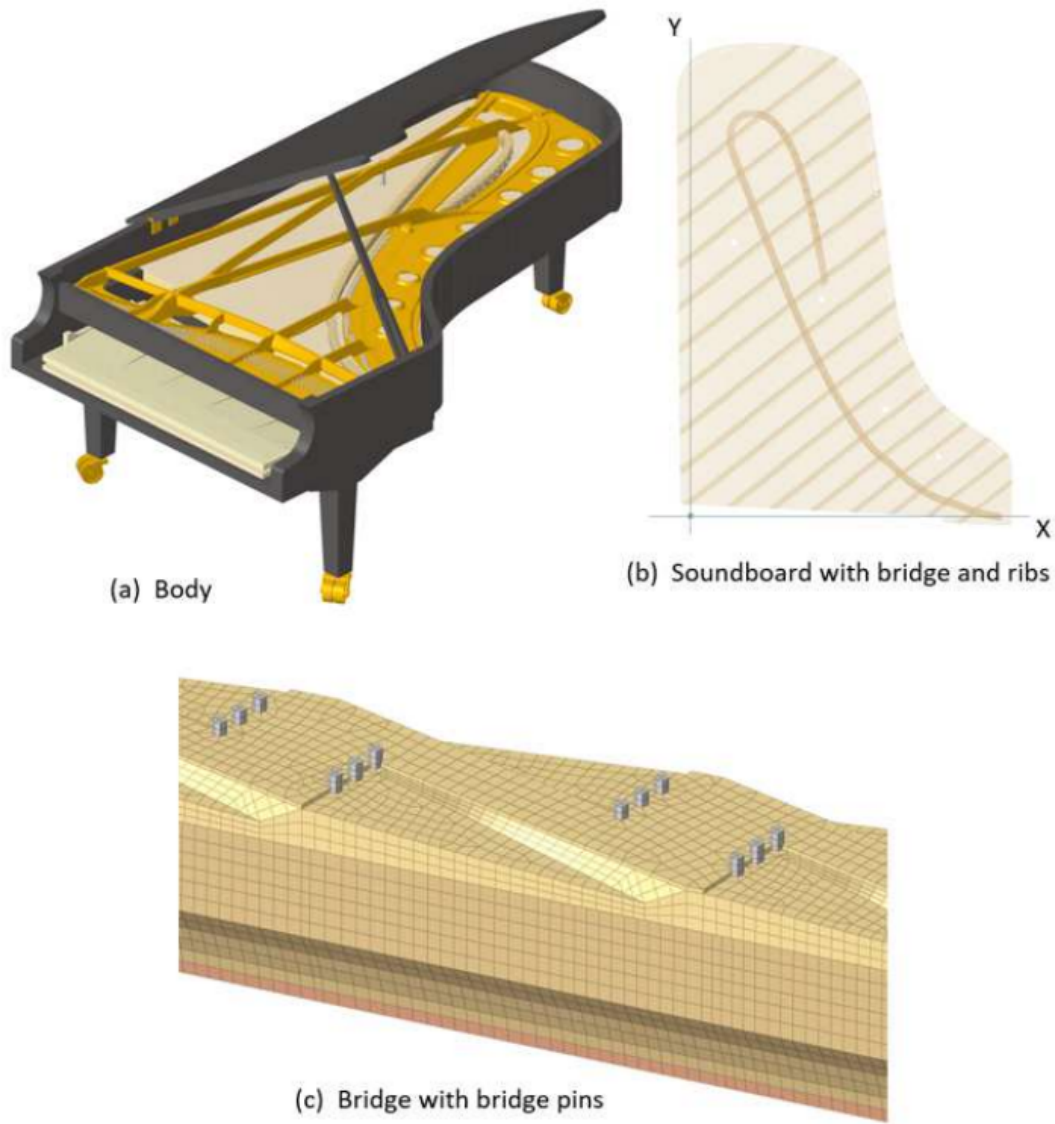


Fig. 8 Three-dimensional finite-element model of the Yamaha concert grand piano CFX.

图 8

Yamaha CFX 音乐会三角钢琴的三维有限元模型。

本页主要为全琴体网格可视化。模型覆盖响板、肋木、琴桥、外壳、铸铁板、琴腿、顶盖等主要结构，并作为制造残余应力、琴体模态及琴体-空气耦合分析的几何基础。

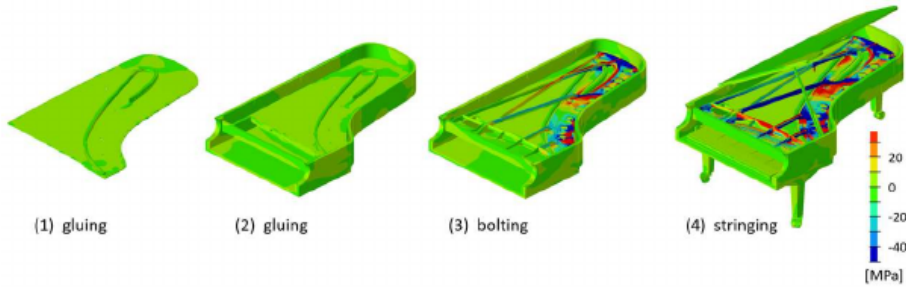


Fig. 9 Residual stress distribution during successive manufacturing processes. The color scale indicates the signed von Mises stress. As expected, a large amount of stress was generated, especially in the cast-iron frame, during the stringing process.

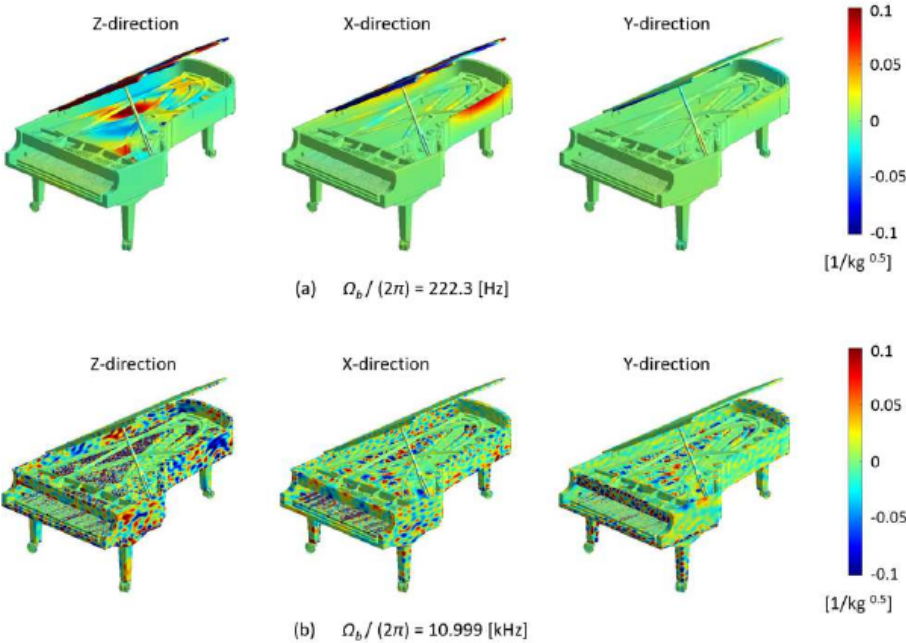


Fig. 10 Representative real normal modes of the piano body obtained by real eigenvalue analysis. Approximately 19,000 real eigenvalues were extracted in the frequency range of 0–11 kHz.

图 9–10

图 9: 连续制造过程中的残余应力分布。颜色标度表示带符号 von Mises 应力。正如预期，上弦阶段尤其在铸铁板中产生很大的应力。

图 10: 琴体实特征值分析得到的代表性实正规模。0–11 kHz 范围提取约 19,000 个实特征值。

这些结果说明本文不是直接把“已完成的琴体”当成无预应力线性结构，而是把制造史带来的几何与应力状态传递到后续模态与声音计算。

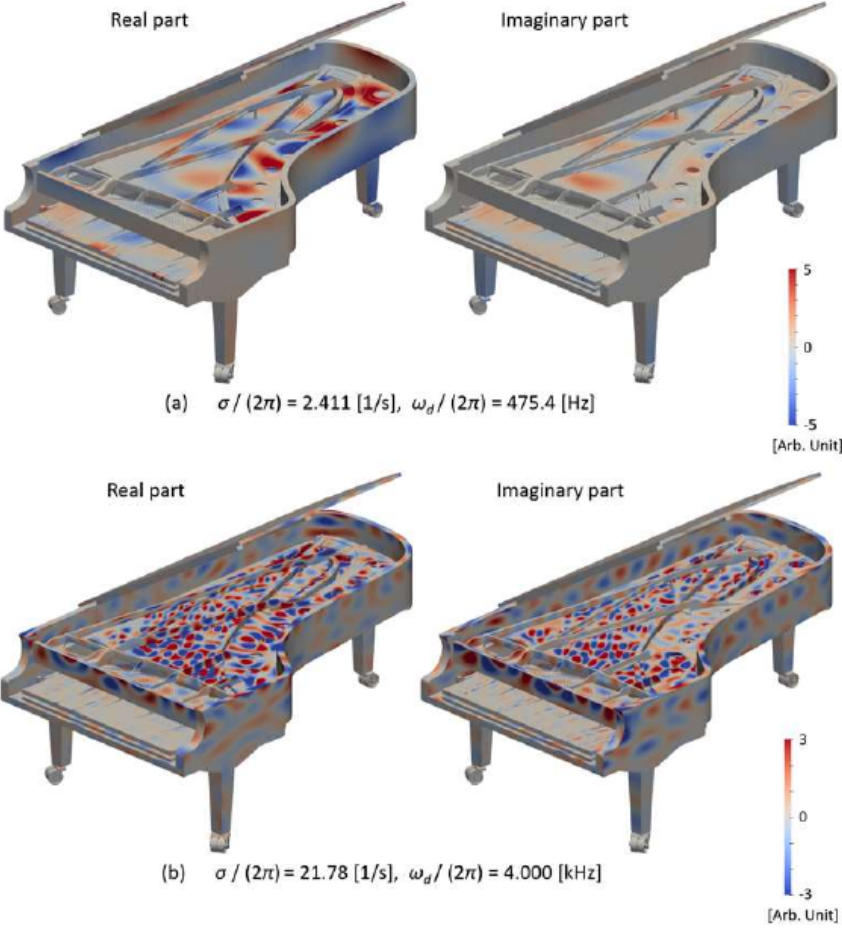


Fig. 11 Representative complex eigenmodes of the coupled body–air system, illustrating normal surface displacement. Approximately 14,000 complex eigenvalues were extracted in the region on the complex plane in the frequency range of 0–5 kHz and the range of the modal decay rate divided by 2π between $0 s^{-1}$ and $120 s^{-1}$.

图 11

琴体-空气耦合系统的代表性复特征模态，图中显示结构表面法向位移。

在 0–5 kHz 的频率范围，并限制模态衰减率 $\sigma / (2\pi)$ 在 $0\text{--}120 s^{-1}$ 内，提取约 14,000 个复特征值。

这些复模态随后被归一化并用于 complex CMS，使大尺度外部振动声学系统可以在时域中由大量一阶复模态状态表示，而无需逐时间步直接求解数百万自由度的 BEM/FEM 系统。

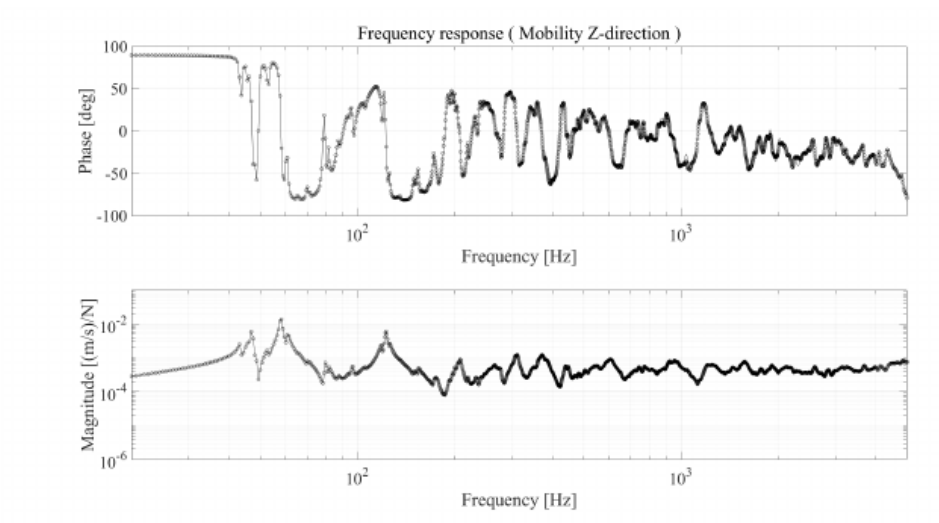


Fig. 12 Driving-point mobility at a string support point on the bridge (Z-direction, note C#3). The solid line represents the directly computed frequency response, and the circles denote the response reconstructed by complex mode synthesis. The circles closely match the solid line.

Figs. 12–14 show the frequency response (driving-point mobility) of the velocity at a string support point when the force is applied at the same string support point. These figures correspond to the bridge in the Z-, X- and Y-directions of note C#3, respectively. The circles (the reconstructed frequency response from the mode synthesis shown by the first term on the right side of Eq. (29)) closely match the solid lines (the frequency response from the frequency response analysis shown by the second term on the right side of Eq. (29)). The curve fitting was satisfied simultaneously at all string support points in the three directions, which indicates that the frequency response was accurately reconstructed by superposing the complex normal modes.

图 12 与驱动点导纳重建

图 12: C#3 琴桥弦支承点 Z 方向的驱动点导纳。实线为直接计算的频率响应，圆点为 complex mode synthesis 重建的响应，两者高度吻合。

图 12–14 分别对应 C#3 琴桥支承点的 Z、X、Y 三方向。当在同一支承点施力时，比较该点速度频响。圆点（由式 29 右侧第一项的模态叠加重建）与实线（式 29 第二项的直接频响分析）在三方向均吻合。

拟合是同时所有弦支承点和三个方向进行的，因此结果表明：通过复正规模叠加，可以高精度重建弦端所“看到”的琴体-空气动态导纳。

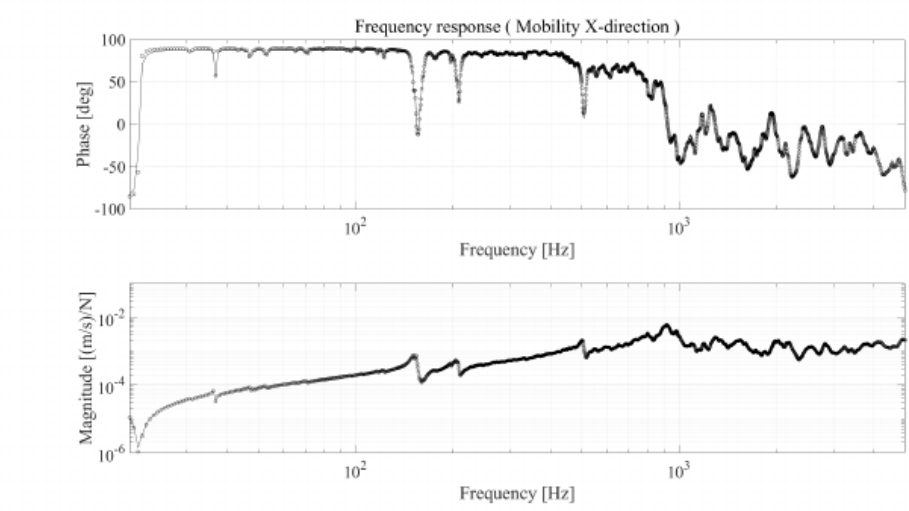


Fig. 13 Driving-point mobility at a string support point on the bridge (*X*-direction, note C#3). The solid line represents the directly computed frequency response, and the circles denote the response reconstructed by complex mode synthesis. The circles closely match the solid line.

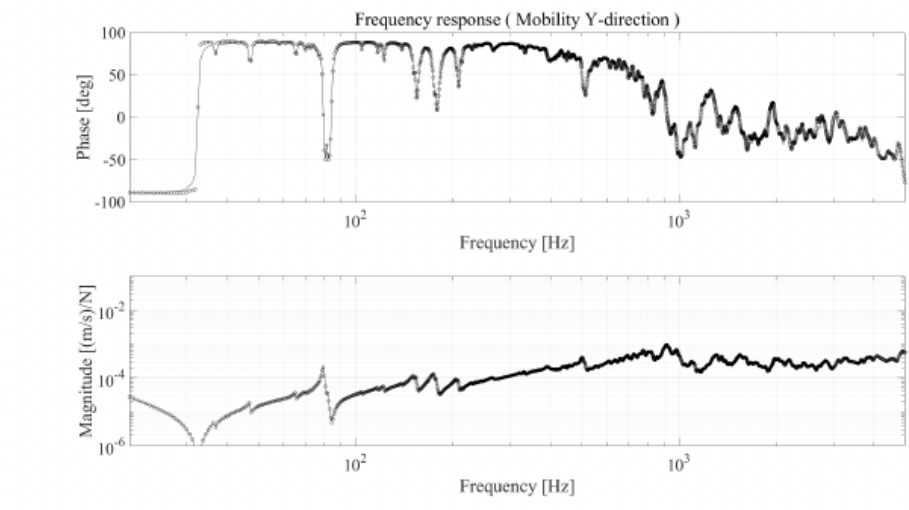


Fig. 14 Driving-point mobility at a string support point on the bridge (*Y*-direction, note C#3). The solid line represents the directly computed frequency response, and the circles denote the response reconstructed by complex mode synthesis. The circles closely match the solid line.

图 13–14

图 13: C#3 琴桥弦支承点 *X* 方向驱动点导纳。实线是直接频响，圆点是复模态综合重建，吻合良好。

图 14: 同一支承点 *Y* 方向驱动点导纳，直接计算与复模态重建同样高度一致。

这两个方向与 *Z* 方向共同验证了三维弦端–琴体接口的复模态降阶质量。

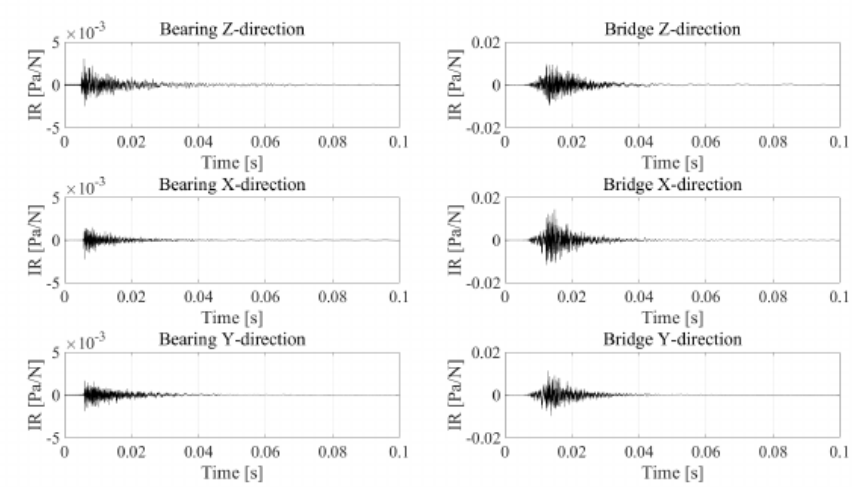


Fig. 15 Impulse responses of sound pressure at a virtual microphone position due to three-dimensional forces applied at the string support points of note E1. Notably, the Z-, X-, and Y-directions refer to global coordinates relative to the body, not the local coordinates of each string.

Fig. 15 shows the impulse response (IR) of the sound pressure at a virtual microphone point when 3D forces are injected in the string support points of note E1. Although the IR of the bearing is smaller than that of the bridge, it should not be ignored when precise sound simulations are required. This figure also shows that the IR of the bearing increases faster than that of the bridge.

Thus, the dynamic characteristics of the large coupled body–air system are reduced to modal parameters (i.e., complex eigenvalues and complex normal modes at the string support points) and impulse response (from the string support points to the virtual microphone points).

4.2 Time-domain simulation

Before evaluating the results of the time-domain simulation, we confirmed that the numerical solution satisfies the energy theorem based on Newton’s laws, which is a necessary condition to indicate that the numerical results were correct, i.e., there were no bugs in the programs.

Notably, herein, the 3D coupled problem between the hammer, the strings, and the body–air system was solved in the time domain. Fig. 16 shows the 3D force exerted by the struck string (the speaking part) at its support points, which is calculated using Eqs. (13a) and (13c). Because the string support point on the bearing is not far from the striking point, the first impulse waveform on the upper left side of Fig. 16 nearly shows the waveform of the force exerted by the hammer on the string. By contrast, because the string support point on the bridge is far from the striking point, the first impulse waveform on the upper right side of Fig. 16 shows the effect of dispersion of

图 15; 4.2 时域仿真

图 15 给出在 E1 的各弦支承点分别施加三维力时，某虚拟麦克风位置的声压脉冲响应。图中的 Z/X/Y 是琴体全局坐标，而非各根弦的局部坐标。bearing（弦枕/铸铁板侧）的 IR 虽然小于琴桥侧，但若追求精确声音模拟不能忽略；其响应上升还更快。

因此，大尺度琴体-空气系统的动态特性最终被压缩成两类数据：弦支承点处的复特征值/复正规模，以及弦支承点到虚拟麦克风的脉冲响应。

4.2 时域仿真开始前，作者首先确认数值解满足基于牛顿定律的能量定理，把它作为程序数值正确、无明显 bug 的必要条件。随后在时域中求解击锤、琴弦和琴体-空气的三维耦合问题。

图 16 将展示 E1 有效弦段在支承点施加的三维力。因为 bearing 离击弦点近，其第一个脉冲几乎就是击锤作用于弦的力波形；琴桥离击弦点远，第一个脉冲会更明显呈现传播色散。

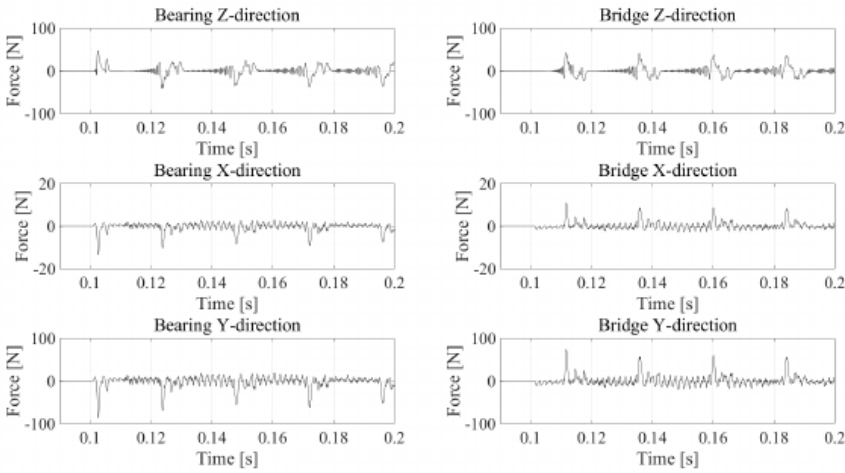


Fig. 16 Time histories of forces exerted by the speaking string of note E1 at its support points following a hammer strike at 4 m/s. At $t = 0.1\text{s}$, the hammer tip is located 3 mm from the string at rest.

the transverse wave more clearly (Podlesak and Lee [35]). The forces on the bridge are transmitted faster in the X - and Y -directions than in the Z -direction because the propagation of the longitudinal wave of the string is faster than that of the transverse wave (see the right side of Fig. 16). Fig. 17 shows the sound pressure at two virtual microphone points. The sound pressure is synthesized by convolving the impulse response from each string support point to each virtual microphone point (shown in Fig. 15) with the force exerted by each string (shown in Fig. 16) and summing that at the ends of the speaking and rear duplex parts of all strings (Eq. (34)). The results in Figs. 16 and 17 were calculated under the condition that the string of note E1 was struck by the hammer at a speed of 4 m/s (a forte playing).

4.3 Piano Sound Synthesis

MIDI (Musical Instrument Digital Interface) data of various musical pieces recorded on a Yamaha Disklavier was used as input for the time-domain simulation, and the piano sounds at the pianist’s listening position were synthesized. To demonstrate the reality of the simulated sounds and the usefulness of the auralization simulation, several audio samples are provided as supplementary materials (Online Resources 1–6).

First, we illustrate the effect of the dispersion of bending waves in the string and the geometrical nonlinearity of the string on the piano sound. The dispersion of bending waves in the string is caused by the string’s bending stiffness (which is the main factor), rotational inertia, and shear deformation. Resources 1–4 are examples

图 16: 弦端力的传播与色散

E1 以 4 m/s 锤速击弦后，有效弦段在两个支承端施加的三维力随时间变化； $t=0.1\text{ s}$ 时锤尖初始位置距静止弦 3 mm。

琴桥侧距离击弦点较远，所以最早到达的横向力脉冲已经显著受到弯曲波色散影响。纵向波传播速度高于横向波，因此与纵向运动关联更强的 X 、 Y 方向力会比 Z 方向更早传至琴桥。

图 17（下一页）给出两个虚拟麦克风位置的声压。它通过把图 16 中各根弦在有效/后弦段两端的三维力，与图 15 的相应脉冲响应卷积并求和得到。图 16、17 的条件均为 E1 锤速 4 m/s，代表 forte。

4.3 钢琴声音合成

作者使用 Yamaha Disklavier 记录的 MIDI 数据作为时域仿真输入，并在演奏者听音位置合成钢琴声，提供 Online Resources 1–6 作为听觉示例。

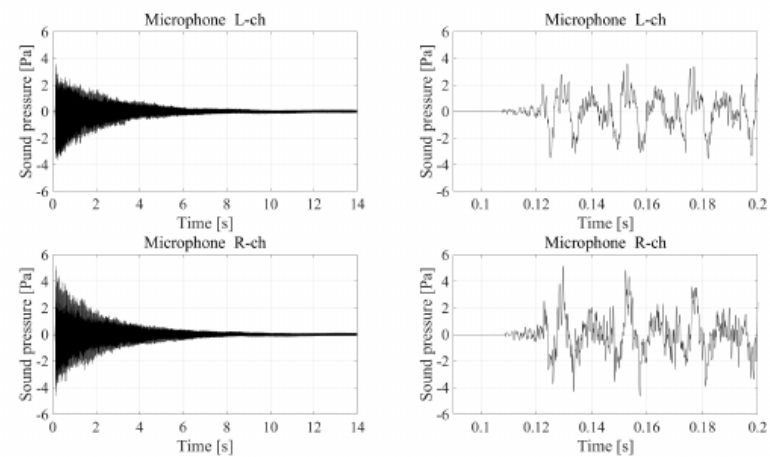


Fig. 17 Synthesized sound pressure waveforms at two virtual microphone positions for note E1 struck at a hammer velocity of 4 m/s. At $t = 0.1\text{s}$, the hammer tip is located 3 mm from the string at rest.

of low range chromatic scale by a hard strike (hammer velocity of 5 m/s). Resource 1 is the recorded sound of Yamaha CFX and Resources 2–4 are the simulated sounds. Resource 2 is the case where both dispersion and geometrical nonlinearity are ignored. This case was calculated as $c'_3 = c'_4 = c'_5 = c'_6 = 0$ in Eq. (6). Resource 3 is the case where dispersion is taken into account and geometrical nonlinearity is ignored. This case was calculated as $c'_3 = 0$ in Eq. (6). Resource 4 is the case where both dispersion and geometrical nonlinearity are taken into account. A comparative listening test among Resources 1–4 clarified that the importance of wave dispersion and geometrical nonlinearity in real piano sound. Resource 1 explains what happens if all partials' phase velocities of the string's bending wave are equal. The metallic ringing sound was clearly audible only in Resources 1 and 4.

The characteristics of the piano sound significantly depend on not only how the piano designer designs but also how the piano technician tunes. The next samples presented as Resource 5 and 6 clarify the effect of the deviation in unison tuning, i.e., the difference in pitch of multiple strings corresponding to one key, on the sound. These cases were calculated by changing ε_0 in Eq. (6). Resource 5 is a sound in which the pitches of multiple strings assigned to the same key are intentionally detuned from one another, within a range that is considered acceptable in practical piano tuning. Specifically, for a three-string unison, the pitch difference between the second string and the first string is set to 0.3 cents in the low range, 1.5 cents in the mid-range, and 2.5 cents in the high range, while the deviation of the third string is half of these values. Resource 6 is a reference sound in which all strings assigned to the same key are tuned to exactly the same pitch, representing an idealized, perfectly aligned condition used for comparison. Here, cent is a unit of pitch difference corresponding to 1/100 of a semitone; for a typical piano string, a deviation of 1 cent corresponds to only a few microns of string stretch. Consequently, this perfectly aligned condition represents an idealized reference that cannot be realized on an actual instrument.

图 17 与色散/几何非线性消融

图 17: E1 以 4 m/s 锤速击弦时，两个虚拟麦克风位置的合成声压波形。

作者首先展示弦的弯曲波色散和几何非线性对声音的影响。色散来自弯曲刚度（主要因素）、转动惯量和剪切变形。Resources 1–4 是低音区半音阶的强击（锤速 5 m/s）：Resource 1 为 Yamaha CFX 实录；Resource 2 同时忽略色散与几何非线性（式 6 中 c'_3, c'_4, c'_5, c'_6 置零）；Resource 3 保留色散但关闭几何非线性（ $c'_3=0$ ）；Resource 4 同时保留色散与几何非线性。

对比听音表明真实钢琴声中波色散和几何非线性都重要；明显的金属 ringing 只在真实录音 Resource 1 与完整模型 Resource 4 中清楚出现。

随后 Resources 5/6 用于比较同音弦调律偏差。Resource 5 给同一琴键多根弦加入实际调律可接受的失谐：三弦同音组中第二根相对第一根在低/中/高音区分别为 0.3/1.5/2.5 cent，第三根偏差为这些值的一半。Resource 6 则把同组所有弦设成完全同频。作者指出完全同频的理想状态在真实钢琴中实际上无法实现。

The characters (a), (b) following Resource No. denote Scarlatti Sonata in E, Chopin Fantaisie-Impromptu, respectively. Resources 5 and 6 demonstrate that variations in unison tuning produce clearly perceptible changes in the resulting sound, although these differences are difficult to describe quantitatively using verbal descriptors alone.

Here, we emphasize the significance of auralization simulation for pianos. First, eliminating or controlling error factors can only be achieved through virtual experiments on a computer, i.e., simulation. When making an actual prototype, there are many error factors, such as material and manufacturing variations. These errors are inherent and cannot be eliminated. Second, by auralizing the simulation results, listeners can directly experience the causal relationship between specifications and the resulting sound with their own ears without the information missing by verbalization. Third, simulation allows us to explore parameter regimes inaccessible to physical prototyping. Resources 2, 3 and 6 are exactly such cases. In other words, simulation is not a simple substitute for the real thing.

In our time-domain simulation, it took about 1,200 seconds to synthesize one sound per second using 88 cores of a CPU (Intel Xeon Platinum 8592). For now, PARTITA still has a lot of room for improvement in parallelization efficiency.

同音弦失谐、可听化意义与计算成本

Resource 编号后的 (a)、(b) 分别代表 Scarlatti 《E 大调奏鸣曲》和 Chopin 《幻想即兴曲》。Resources 5 与 6 表明，同音弦微小调律差异会造成清楚可听的音色变化，但这种差异很难单靠语言定量描述。

作者强调钢琴可听化仿真的意义：第一，只有计算机虚拟实验才能完全消除或控制材料/制造误差变量，真实样机无法做到；第二，把仿真结果直接变成声音，可让听者不经语言压缩就“听到”规格变化与声音变化之间的因果关系；第三，仿真可以探索真实样机不可实现的参数状态，Resources 2、3、6 就属于这种情况。因此仿真并非只是实物的替代品。

当前实现计算量极大：使用 Intel Xeon Platinum 8592 的 88 个 CPU 核，合成 1 秒声音约需 1200 秒。作者指出 PARTITA 的并行效率仍有大量优化空间。

5 Discussion

The primary objective of this work is not parameter identification accuracy, but the demonstration of a causal, integrable simulation framework. The presented framework numerically reproduces hallmark phenomena of piano sound production within a single, causally consistent simulation loop that spans hammer felt, spatial strings, and a 3D body–air system. In particular, the model captures the experimentally reported exchange of energy between the two transverse string polarizations (often termed “double polarization”) and the emergence of metallic, non harmonic “ringing.” Figs. 18–21 show that the simulated three-dimensional displacement and trajectories at the striking point evolve from predominantly vertical to increasingly horizontal motion, with elliptical loops and rotation direction changes comparable to reported measurements [36], thereby supporting the physical fidelity of the formulation. The key point is that it precisely expresses the three-dimensional connection between the strings and the body. By explicitly modeling the zig-zag end condition (inclined bridge pins) together with rotational spring support and distinct normal/tangential stiffnesses at both supports, the simulation naturally reproduces cross polarization coupling without resorting to ad hoc terms. The bidirectional coupling between longitudinal and transverse motion—retained to satisfy energy conservation—further explains amplitude dependent spectral enrichment and conditions under which partial sums lock to longitudinal resonances, contributing to audible ringing.

On the hammer side, a double layer nonlinear generalized Maxwell model was introduced to account for disparate viscoelastic behaviors of the tip and core. While Section 2 identified parameters via differential evolution based on the sound, we note that direct validation against measured force–compression loops and time resolved hammer–string force would further strengthen the claim of wide range fidelity. Such comparisons will quantify hysteresis, rate dependence, and the transition between soft and strong strikes across keys.

Regarding the body–air subsystem, the complex mode CMS based on nonlinear eigenvalue analysis reconstructs driving point mobilities (Figs. 12–14) with high accuracy across three spatial directions at the supports, demonstrating that dense exterior vibroacoustics can be reduced to a compact set of complex eigenvalues and residues suitable for time-domain use. In our current implementation, to contain cost we neglect air to structure feedback above 5kHz. This approximation is justified by the relatively small added mass and radiation induced stiffness corrections at high frequency compared with internal losses and by the fact that auralization in this range is dominated by structure to air radiation already captured by the identified modes. As computational resources improve, we plan to remove this high frequency simplification.

From an engineering science perspective, two aspects are broadly transferable. First, the manufacturing aware modeling: residual stresses introduced by gluing, bolting, and stringing measurably shift natural frequencies and modal damping. Embedding nonlinear static assembly analysis into the acoustic pipeline enables causal, design for sound studies not only for pianos but for other large wooden or composite structures where anisotropy and residual stress are significant. Second, the CMS centric reduction: combining hybrid interface CMS for nonlinear substructures

5 讨论

本研究首要目标并非参数识别精度，而是证明一个因果一致、可统一积分的仿真框架。它在同一个闭环中跨越击锤毡、空间琴弦和三维琴体–空气系统，并重现钢琴发声的标志性现象。尤其是，模型捕捉到实验报道过的两个横向偏振之间的能量交换（double polarization），以及金属、非谐的 ringing。

图 18–21 显示，E1 击弦点三维位移轨迹从主要竖直运动逐渐增加水平分量，形成椭圆环，并发生旋转方向改变；与既有实验测量相似，支持该模型的物理可信度。关键在于它精确表达弦与琴体之间的三维连接：倾斜桥钉产生的 zig-zag 条件、转动弹簧支承、两端不同的法向/切向刚度，使交叉偏振耦合自然产生，而不需要人为经验耦合项。

为满足能量守恒而保留的纵向–横向双向耦合，还可解释随振幅增加的频谱丰富化，以及横向分音和频率与纵向共振锁定时出现的可听 ringing。

击锤方面，双层非线性广义 Maxwell 模型用于区分锤尖和核心不同的黏弹行为。不过第 2 节参数是通过基于声音的差分进化识别的；若能直接与测得的力–压缩迟滞环和高时间分辨率 hammer–string force 比较，将更有力验证其跨力度范围真实性。

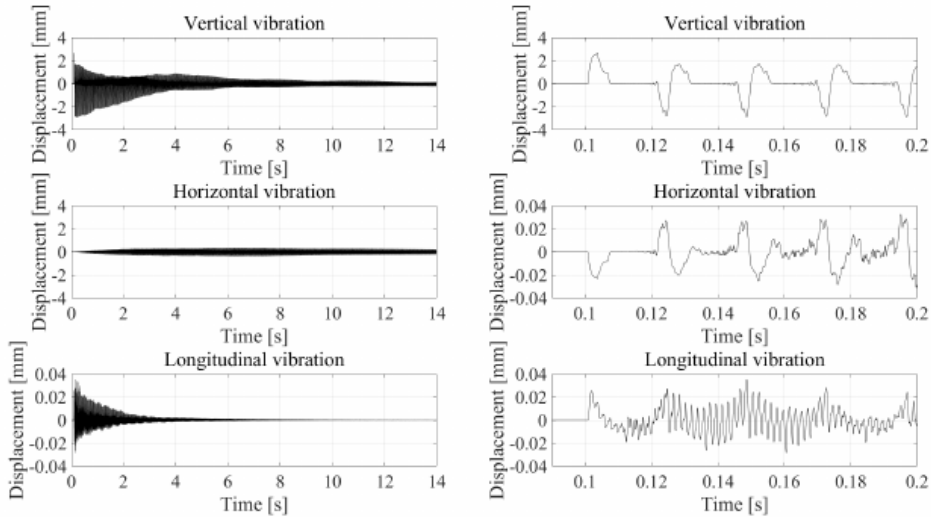


Fig. 18 Three-dimensional displacement of the string at the striking point of note E1 following a hammer impact at 4 m/s. At $t = 0.1$ s, the hammer tip is located 3 mm from the string at rest. The vertical, horizontal, and longitudinal vibrations correspond to z , inverted y , and x -directions in the local coordinates of the string, respectively.

with complex mode synthesis for exterior vibroacoustics provides a scalable route to parameter sensitive, time-domain simulations without sacrificing key perceptual cues. Limitations suggest clear opportunities. We exclude action mechanism dynamics and damper behavior from the time-domain loop; incorporating these will refine onset/decay envelopes and pedaling effects. Finally, while we provided aural examples distinguishing dispersion and geometric nonlinearity and illustrating unison detuning, a systematic perceptual study (e.g., MUSHRA) would quantify thresholds and design sensitivities. Overall, the results indicate that auralization driven engineering—tight coupling of high fidelity modeling, manufacturing realism, and perceptual evaluation—can make specification level choices (e.g., scale design, wood properties, crown geometry) and technician level adjustments (e.g., unison tuning) audibly and reproducibly traceable. We anticipate the method will accelerate evidence based piano design and inform scalable, physics based sound synthesis in other complex vibroacoustic systems.

图 18 与讨论 (续)

图 18: E1 在 4 m/s 击锤冲击后, 击弦点的三维位移。t=0.1 s 时锤尖初始位置距静止弦 3 mm。弦局部坐标中, z 对应竖直振动, $-y$ 对应水平振动, x 对应纵向振动。

琴体-空气方面, complex CMS 基于非线性特征值分析, 在三个空间方向上都高精度重建弦端驱动点导纳。这说明稠密的外部振动声学系统可以被压缩为适用于时域的复特征值和残数集合。当前为降低成本, 5 kHz 以上忽略空气→结构反馈; 作者认为高频附加质量/辐射刚度相对内部损耗较小, 且结构→空气辐射仍被保留, 因此该近似目前合理, 未来将取消。

从工程科学角度, 两点具有普遍性: 一是把胶合、螺栓和上弦产生的制造残余应力嵌入声学仿真; 二是以 CMS 为核心, 把非线性子结构的混合界面 CMS 与外部声学的复模态综合结合, 从而在不丢失关键听觉线索的前提下进行参数敏感的时域仿真。

局限: 动作机构动力学与真实制音器行为尚未完整进入时域闭环; 当前可听化是示例性质, 还缺少 MUSHRA 等系统感知实验。

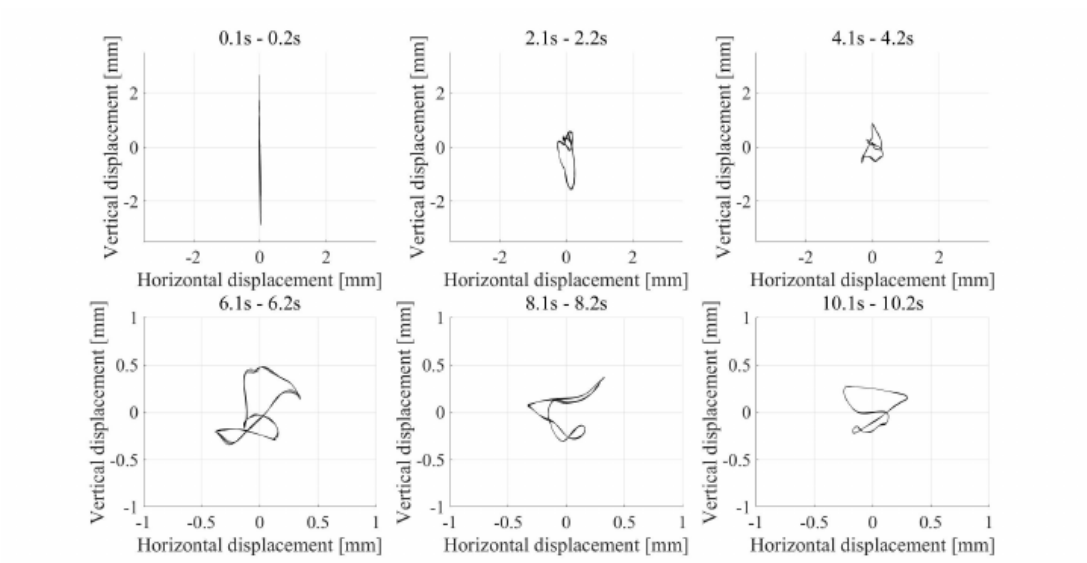


Fig. 19 Trajectories consisting of the components in the z -direction and inverted y -direction of the displacement shown in Fig. 18

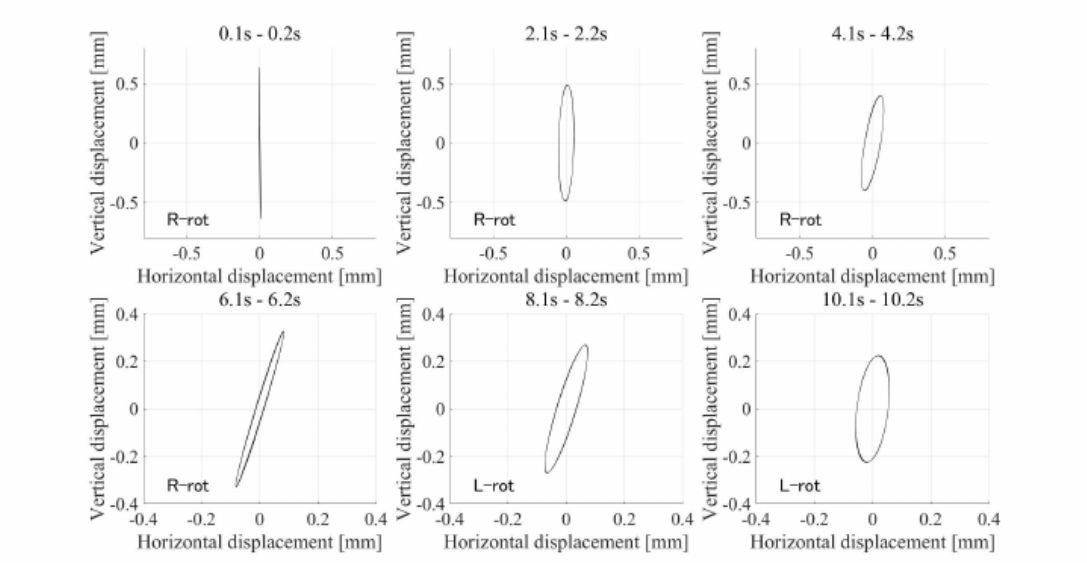


Fig. 20 Trajectories of the fundamental of the displacement shown in Fig. 19. R-rot and L-rot indicate right and left rotations from the pianist's listening point, respectively.

图 19–20

图 19: 由图 18 位移的 z 分量和反向 y 分量构成的二维轨迹。可直接看到弦振动轨迹从近似单轴逐渐变成椭圆，并发生偏振能量交换。

图 20: 图 19 中“基频分量”的轨迹。R-rot 与 L-rot 分别表示从演奏者听音位置观察的右旋与左旋。这两图用于展示弦的双偏振并不是简单固定椭圆，而会随时间发生相位、能量分配与旋转方向的演化。

图 21

图 21：图 19 中“二阶分量”的轨迹。

与基频轨迹对照可见，不同阶次分音在两个横向偏振上的能量分配和相位关系并不相同。这正是端部各向异性与三维耦合在模态层面留下的动态特征。

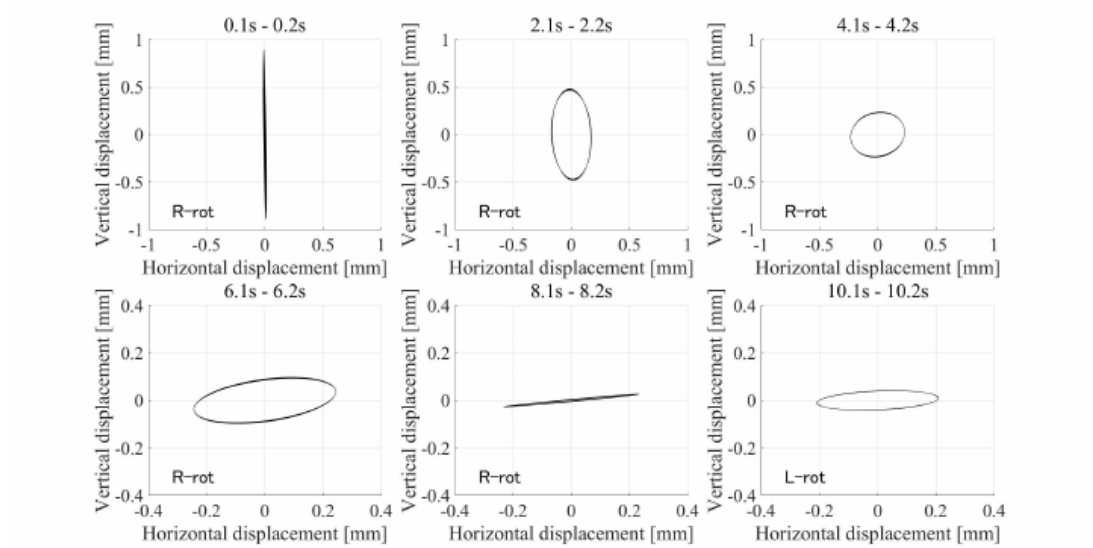


Fig. 21 Trajectories of the second-order component shown in Fig. 19.

6 Conclusions

This paper introduced a physics-based, time-domain simulation framework that—unlike prior studies focusing on isolated subsystems—treats the piano as a fully integrated, three-dimensional coupled system. The model incorporates nonlinear hammer–string interaction, geometrically nonlinear string vibration with end-support anisotropy, the orthotropic complex elasticity of wood, and exterior vibroacoustics via a large-scale body–air model. By combining hybrid-interface CMS + Galerkin method for strings with FEM + FMBEM and a complex CMS based on nonlinear eigenvalue analysis for the body–air subsystem, the approach achieves substantial model-order reduction while preserving the essential dynamics required for perceptually faithful sound synthesis.

Primary contributions are threefold.

- (i) System-level integration in the time domain: We simulate the complete hammer–string–body–air chain, enabling causal analysis from design parameters and manufacturing stress to radiated sound pressure.
- (ii) Mechanism-faithful modeling of nonlinearities and anisotropy: A double-layer nonlinear generalized Maxwell model captures the felt hysteresis across a wide dynamic range, while geometrical nonlinearity and zig-zag/anisotropic boundary conditions at the string supports reproduce dispersion, polarization exchange, and ringing with high fidelity.
- (iii) Scalable reduction for exterior vibroacoustics: The combination of FMBEM and complex CMS efficiently handles a multi-million-DOF coupled body–air problem, yielding complex modal parameters at the string support points and impulse responses to microphone positions for accurate auralization.

Perceptual validity was supported through audible examples that reconstruct salient piano phenomena—ringing, partial dispersion, and unison detuning—within a single unified physical model. These results indicate that the proposed framework can quantitatively connect specification-level choices (e.g., string support conditions, wood properties, residual stresses) and technician-level adjustments (e.g., unison tuning) to audible outcomes. As such, it provides a practical foundation for auralization-based design exploration, pedagogy, and virtual prototyping.

Limitations and scope. The present implementation neglects action mechanics in the time domain and, in the high-frequency band, ignores the feedback force exerted by air onto the body to reduce computational cost. These pragmatic choices do not alter the main conclusions but suggest clear opportunities for refinement.

Future directions. We plan to:

- (i) incorporate action-mechanism dynamics (including key–hammer transfer characteristics and damper behavior) [37] into the time-domain loop;
- (ii) remove the high-frequency air-to-structure feedback approximation as computational resources advance;
- (iii) tighten parameter identification (e.g., loss factors, boundary anisotropy, relaxation times) via Bayesian or differential-evolution fitting against multi-modal measurements;

6 结论

本文提出基于物理的时域仿真框架，与过去只研究孤立子系统的方法不同，它把钢琴作为完整的三维耦合系统。模型同时包含：非线性击锤–弦相互作用；具有端部各向异性和几何非线性的弦振动；木材正交各向异性复弹性；以及大尺度琴体–空气外部振动声学。琴弦使用混合界面 CMS + Galerkin；琴体–空气使用 FEM + FMBEM，再通过基于非线性特征值分析的 complex CMS 大幅降阶，同时保留感知上重要的动力学。

主要贡献三点：

- (i) 时域系统级整合：模拟完整 hammer–string–body–air 链，从设计参数与制造应力一路因果追踪到辐射声压。
- (ii) 对非线性与各向异性进行机理忠实建模：双层非线性广义 Maxwell 模型描述宽力度范围的毡迟滞；弦的几何非线性及 zig-zag/各向异性支承高保真重现色散、偏振交换与 ringing。
- (iii) 面向外部振动声学的可扩展降阶：FMBEM + complex CMS 处理数百万自由度耦合系统，输出弦支承点的复模态参数以及到麦克风位置的脉冲响应。

可听示例从统一模型中重现 ringing、分音色散和同音弦失谐，支持其感知有效性。

(iv) extend the method to other stringed instruments (e.g., guitar, violin) by reusing the same CMS-centric pipeline; and

(v) investigate real-time implementations (reduction of elements that are not auditorily dominant, maximizing parallel efficiency of the program, etc.) toward next-generation, physics-based tone generators.

In summary, this study demonstrates a numerical time-domain realization of a full-scale grand piano as a comprehensive coupled physical system. The proposed framework bridges the gap between mechanism-level modeling and perceptual evaluation, enabling reproducible, parameter-sensitive experiments that are impractical with physical prototypes alone. We anticipate that this approach will accelerate evidence-driven piano design and inform high-expressivity sound synthesis beyond conventional PCM-based methods.

Acknowledgements. The authors thank their colleagues at Yamaha who contributed to this study. The eigenvalue and nonlinear static analysis methods for the large-scale model using MSC Nastran were developed by Hitoshi Izutani and Makoto Minoda. Most of the Yamaha CFX measurements were performed by Junji Kuroda. A part of the signal processing algorithm in PARTITA was implemented by Onsai Ryu. The authors thank Masatsugu Okazaki for the helpful discussions on this research. One of the authors, Eiji Tominaga, thanks Takashi Tamaki, who wrote the in-house reports on experimental research into the ringing sound in 1984, and Hajime Hayashida, the former supervisor, for the opportunity to get started this research.

Data availability. The datasets generated and analyzed during the current study are not publicly available due to their size and proprietary restrictions, but are available from the corresponding author upon reasonable request.

Supplementary materials. Audio files (WAV, 44.1kHz, 24-bit, stereo) associated with this article are provided as Electronic Supplementary Material (Online Resources 1–6).

- Online Resource 1: Low range chromatic scale (Recorded sound of Yamaha CFX)
- Online Resource 2: Low range chromatic scale (Simulated sound in the case where both dispersion and geometrical nonlinearity are ignored.)
- Online Resource 3: Low range chromatic scale (Simulated sound in the case where dispersion is taken into account and geometrical nonlinearity is ignored.)
- Online Resource 4: Low range chromatic scale (Simulated sound in the case where both dispersion and geometrical nonlinearity are taken into account.)
- Online Resource 5(a): Scarlatti Sonata in E (Maximum mistuning: 2.5cent)
- Online Resource 6(a): Scarlatti Sonata in E (Perfect tuning)
- Online Resource 5(b): Chopin Fantaisie-Impromptu (Maximum mistuning: 2.5cent)
- Online Resource 6(b): Chopin Fantaisie-Impromptu (Perfect tuning)

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当前实现的主要简化：时域中忽略动作机构动力学；高频段为降低计算量忽略空气对琴体的反馈力。这些工程折中不改变主要结论，但为后续改进留下空间。

未来计划：

(i) 把琴键–击锤传递特性与制音器行为等动作机构动力学加入时域闭环；

(ii) 算力提高后取消高频空气→结构反馈近似；

(iii) 通过 Bayesian 或差分进化，结合多模态测量，更严格识别损耗因子、端部各向异性和松弛时间等参数；

(iv) 复用 CMS 核心流程扩展到吉他、小提琴等其他弦乐器；

(v) 研究实时实现：删减听觉上不占主导的元素，并最大化程序并行效率，面向下一代基于物理的音源。

总之，本研究实现了全尺寸三角钢琴作为完整耦合物理系统的数值时域仿真，把机理级建模与感知评价连接起来，使真实样机难以进行的、可重复且参数敏感的实验成为可能。

致谢与数据：作者感谢 Yamaha 同事参与大型模型、测量与 PARTITA 算法开发。数据集因规模和专有约束未公开，但可向通讯作者合理申请。补充材料为 44.1 kHz / 24-bit / stereo WAV。

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[35] Podlesak & Lee: Dispersion of waves in piano strings (JASA, 1988) ——本文讨论弦色散的重要经典来源。

[36] Tanaka, Nagai, Mizutani: 钢琴弦二维振动测量 (2000, 日文) ——用于讨论双偏振轨迹与实验的一致性。

[37] Tominaga & Sato: 钢琴触键的物理建模与仿真 (2017, 日文) 。

Supplementary Files

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- [OnlineResource2.wav](#)
- [OnlineResource3.wav](#)
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- [OnlineResource5b.wav](#)
- [OnlineResource5a.wav](#)
- [OnlineResource6a.wav](#)
- [OnlineResource6b.wav](#)

补充文件

该预印本附带以下电子补充 WAV 文件：

OnlineResource1.wav：Yamaha CFX 低音区半音阶实录。

OnlineResource2.wav：低音区半音阶仿真，同时忽略色散与几何非线性。

OnlineResource3.wav：保留色散、忽略几何非线性。

OnlineResource4.wav：同时保留色散与几何非线性。

OnlineResource5a.wav / 6a.wav：Scarlatti 《E 大调奏鸣曲》，分别为有实际同音弦失谐与完全同调。

OnlineResource5b.wav / 6b.wav：Chopin 《幻想即兴曲》，同上两种调律条件。